
■ READING UPDATE – 2026-09-16. No value in this paper is changed by this notice.
How the corpus is now read: One coordinate up – <https://osf.io/ujwc7>
DOI 10.17605/OSF.IO/UJWC7

UPDATE NOTICE – 16 September 2026

This reprint adds the three lines above this notice (a pointer to One coordinate up, where the corpus is now read) and carries two small corrections and one erratum; no identity, bound, fit or measured value changes.

- (1) Three lines added under the title – READING UPDATE – pointing to One coordinate up (<https://osf.io/ujwc7> · DOI 10.17605/OSF.IO/UJWC7), the synthesis where the corpus is now read. The paper is classed as a NOTE on the Observational Record shelf of the framework's Direct Documents page, under the short title "The η Corridor"; the Verification Companion is reprinted with it.
- (2) PRINTED-DIGIT CORRECTION, §5.1, SAW row: $2G - \eta$ printed as +0.008441; with $2G = 0.0394837\dots$ and $\eta = 0.0310434$ the difference is +0.008440 (the printed figure used $2G$ rounded to 0.039484). The σ -margin $+2813\sigma$ is unchanged; the other nine rows reproduce.
- (3) ERRATUM, §9.11: the bridge sentence "the Bernoulli prefactor $|B_{(2n)}| \cdot 2^n/n$ equals the reduced tangent number a_n (OEIS A002105)" omits a factor: $A002105(n) = (2^{(2n)} - 1) \cdot |B_{(2n)}| \cdot 2^n / n$. Both results it connects – Theorem 9.7 and Corollary 9.9 – are correct and agree for all n ; a reader who reconstructs the factorization from the sentence as printed carries an extra $(2^{(2n)} - 1)$ and would wrongly conclude the two disagree. Found 2026-08-26 while working on another paper (D.B.), verified at exact arithmetic the same hour (Opus, Cowork), parked for this paper's turn; a bracket in place says it. Integer Mellin Moments of the Dome states the relation correctly. No computed value and no theorem is affected.
- (4) LABEL CORRECTION, §13 convention note: "(the $\sqrt{2}/2$ factor in the 120-cell unit-edge circumradius $R = (\sqrt{2}/2) \cdot \sqrt{(5+\sqrt{5})} \approx 1.9021$)" – that quantity is not the 120-cell's unit-edge circumradius, which is $\sqrt{2} \cdot \varphi^2 \approx 3.7025$ (from the 600 vertex coordinates); it is $2 \sin(2\pi/5) = \sqrt{(\varphi^2 + 1)}$, twice the unit-edge icosahedron circumradius quoted in the same sentence. A bracket in place says so; the note's own conclusion – that the overflow reading is framework-internal and would not go through under a uniform unit-edge convention – already stands and is unaffected. The same label is corrected in Geometric Constants v2 and the Complete Framework.

Every displayed value re-computed 2026-09-11 at 40–90 digits ($2G$, the ten σ -margins, the dome landmarks $g(\sqrt{2})$, $g(\varphi) = \varphi^{(-5/2)}$, $g(2) = 1/(2\sqrt{3})$, the peak at φ , $J(2) = \ln(2)/2 - \pi^2/32$, $R(\text{Ising})$ against $2K_{\text{AUD}}$, and every table of the Verification Companion): all stand. Prior version remains on OSF in the project's archive/ folder, named by its date.

Part I – The Observation

1. Abstract

We report that the anomalous dimension η at $d = 3$ critical points in the Wilson-Fisher fixed-point family – unitary CFTs with positive ν , standard hyperscaling, and a continuous finite-T transition – is empirically bounded above by $2G \approx 0.039484$, where $G = 1 - \sqrt{2} \cdot \ln 2 \approx 0.019742$. The constant $K_{\text{AUD}} = \sqrt{2} \cdot \ln 2 \approx 0.080258$ is the central constant of an underlying framework whose derivation is given elsewhere (Paper 3 V3, DOI: 10.17605/OSF.IO/QH5S2; direct: osf.io/qh5s2). Part I of this paper tests the bound empirically; Part II develops the dome-centric structural reading.

The bound is tested against ten WF-family universality classes – $O(N)$ for $N = 0-4$, cubic anisotropy at $N = 3, 4$, MN cubic at $N = 2, 3$, and randomly dilute Ising – and holds in every case, with σ -margins ranging from $+2.2\sigma$ (randomly dilute Ising, the tightest unitary case) to $+1593\sigma$ (3D Ising, conformal bootstrap). Three classes outside WF-family scope have $\eta > 2G$ but are excluded by independent physical structure: RFIM ($T = 0$ fixed point with modified hyperscaling), deconfined $SU(2)$ QCP (pseudo-critical), and disordered Potts large- q (disorder-induced second order).

The bound is offered as an observation, not a derived result. The priority falsification target is a higher-precision determination of η at $d = 3$ randomly dilute Ising: a factor-of-three improvement would place $\eta = 2G$ at the edge of measurement.

2. Introduction

Scope of this paper. This paper presents (i) an empirical observation: that the anomalous dimension η at $d = 3$ Wilson-Fisher critical points is bounded above by $2G \approx 0.039484$ across all ten universality classes tested; (ii) a mathematical architecture (Part II) – the dome function and its algebraic landmarks – that gives the bound a specific structural shape; and (iii) a Tier-5 conjecture explicitly flagged as conjectural – the Central Synchronization Conjecture, §11, with the framework's six-tier evidence system introduced in §14.4. The paper does not claim a derivation of the bound from first principles, a physical mechanism explaining why K_{AUD} appears across multiple domains, a connection to consciousness, gravity, electromagnetism, fusion, or any other physical theory not explicitly developed here, or that the framework is unique or final. The bound is offered as a falsifiable empirical observation; the structural reading is offered as a coherent organization of what the observation invites, not as a derivation of what the observation requires. A reader may adopt the empirical observation (§§1–8) without committing to the structural reading (Part II); the three reading positions in §14.6 make this nesting explicit.

2.1 The anomalous dimension η

In a continuous phase transition the two-point correlation function at the critical point decays as a power law

$$G(r) \sim 1 / r^{(d - 2 + \eta)}$$

where d is the spatial dimension and η is the anomalous dimension, a universal critical exponent characterizing the universality class of the transition. The combination $2 - \eta$ is the effective correlation dimension appearing in Fisher's exact relation $\gamma = \nu(2 - \eta)$, where γ is the susceptibility exponent and ν is the correlation-length exponent. At the upper critical dimension $d_c = 4$ of ϕ^4 theory (Wilson 1971; Wilson and Fisher 1972), η vanishes exactly and the transition is mean-field. Below d_c the order-parameter fluctuations are strong enough to give η a nonzero value; in $d = 3$ the Wilson-Fisher fixed point produces $O(N)$ universality classes whose η values are accurately measured by conformal bootstrap (Chester et al. 2020, Kos et al. 2016), Monte Carlo simulation, and ϵ -expansion resummation.

The set of measured η values at $d = 3$ is small in number (a few tens of distinct universality classes have been characterized at sub-percent precision) and bounded above by values well below 0.05. The exact upper bound on η at $d = 3$ is not known from first principles. No bound derivable from current conformal-bootstrap technology, RG theory, or hyperscaling relations places η below 0.05.

2.2 The observation

The geometric-informational constant $K_{\text{AUD}} = \sqrt{2} \cdot \ln 2 \approx 0.980258$ appears as the central constant of a framework whose derivation, motivation, and broader architecture are given in Paper 3 V3 (DOI: 10.17605/OSF.IO/QH552). The same value appears independently as the threshold coefficient a in an upper bound $\Delta^*(x) \leq C \cdot \exp(a \cdot \sqrt{(\log_2 x)})$ on the (weighted) average order of the Erdős-Hooley Δ -function, due to de la Bretèche and Tenenbaum (arXiv:2210.13897); the value $\sqrt{2} \cdot \log 2 \approx 0.980258$ is identified there as the threshold coefficient a in that exponential factor. The complement $G = 1 - K_{\text{AUD}} \approx 0.019742$ plays the role of a gap in the framework's architecture.

The doubled gap $2G \approx 0.039484$ is the empirical ceiling on η we report in this paper: across ten known WF-family universality classes at $d = 3$, every measured η is bounded above by $2G$. The framework's derivation of why K_{AUD} specifically (and therefore $2G$ specifically, via Fisher's relation $\gamma = \nu(2 - \eta)$) might bound η is the open theoretical problem; for the present paper we report only the empirical claim. In the language of Fisher's relation the bound is equivalent to $(2 - \eta) \geq 2K_{\text{AUD}} = 2\sqrt{2} \cdot \ln 2 \approx 1.960516$, so the effective correlation dimension of any WF-family critical point in 3D is confined to the corridor $[2K_{\text{AUD}}, 2]$ of width $2G$.

The paper's contribution is the following:

1. The bound holds across all WF-family universality classes in $d = 3$ we have located in the published literature (ten classes: $O(N)$ for $N = 0-4$, cubic anisotropy at $N = 3$ and $N = 4$, MN cubic at $N = 2$ and $N = 3$, and randomly dilute Ising). Five of these are non- $O(N)$ classes that were not previously tested in this framing.
2. The bound is sharply violated outside WF-family scope – by RFIM, DQCP-SU(2), and disordered Potts large- q – but each excluded class has independent, physics-tier reasons for being outside the bound's scope ($T = 0$ fixed point, pseudo-criticality, disorder-induced second order). The exclusions are not ad-hoc.
3. The bound has a concrete priority falsification target (random-bond Ising at $d = 3$ at improved precision), and four further falsification routes are listed in §7.

2.3 What this paper is not

This paper is not a derivation. It is an observation, tested against the published critical-exponent data. The framework's interpretation of K_{AUD} as a geometric-informational intersection lies behind the choice of K_{AUD} as the bound-setting constant, but the bound's empirical status is independent of whether one accepts the framework's structural reading. A reader who accepts the empirical evidence in §5 but rejects the framework's interpretation in Paper 3 V3 may still adopt the bound as a numerically interesting observation about WF-family 3D η values. We are explicit throughout about which claims are observational and which are framework-interpretive.

3. The K_{AUD} architecture (background)

This section briefly summarizes the constants used in this paper; the full derivation is in Paper 3 V3.

Symbol	Value	Definition
K_{AUD}	$\sqrt{2} \cdot \ln 2 \approx 0.980258$	Framework constant: $\sqrt{2}$ (geometric primitive for H_4) $\times \ln 2$ (binary unit).
G	$1 - K_{\text{AUD}} \approx 0.019742$	Framework gap; complement of K_{AUD} relative to unity.
$2G$	$2(1 - K_{\text{AUD}}) \approx 0.039484$	Empirical η ceiling. Algebraically = corridor width $2 - 2K_{\text{AUD}}$.
$2K_{\text{AUD}}$	$2\sqrt{2} \cdot \ln 2 \approx 1.960516$	Conjectured lower bound on $(2 - \eta)$; corridor floor.

Note on the "geometric primitive" usage. The $\sqrt{2}$ in the K_{AUD} row above is the framework's geometric primitive associated with H_4 , not the unit-edge 120-cell circumradius (which is $(\sqrt{2}/2) \cdot \sqrt{(5+\sqrt{5})} \approx 1.9021$). Paper 3 V3 §3.4 catalogues four origins for the $\sqrt{2}$ factor in H_4 geometry; §13.3 of the present paper carries the full convention discussion. The same usage applies to " H_3 circumradius" elsewhere in this paper, where ϕ is the framework's primitive for H_3 rather than the unit-edge icosahedron circumradius 0.9511.

The derivation of K_{AUD} from H_4 polytope geometry and binary distinction is given in:

- D.B., $\sqrt{2} \times \ln(2)$: Geometric Constants from H_4 , Complete Framework V3, DOI: 10.17605/OSF.IO/QH5S2; OSF: osf.io/qh5s2 (project page serves the current version, currently 3.3.1); GitHub raw text (recommended for AI readers): [view raw](https://raw.githubusercontent.com/gap-geometry/gap-geometry.github.io/sqrt2-ln2-geometric-constants-ai-readers.html); framework AI Readers hub: gap-geometry.github.io/sqrt2-ln2-geometric-constants-ai-readers.html. (The OSF project title carries the original "V3.0" designation; the file content has been updated through version 3.3.1, available at the GitHub raw and OSF current-file links.)

The same numerical value appears independently in analytic number theory:

- R. de la Bretèche and G. Tenenbaum, "Two upper bounds for the Erdős-Hooley Delta-function" (2022), [arXiv:2210.13897](https://arxiv.org/abs/2210.13897). The value $\sqrt{2} \cdot \log 2 \approx 0.980258$ appears as the threshold coefficient a in an upper bound $\Delta^+(x) \leq C \cdot \exp(a \cdot \sqrt{(\log_2 x)})$ on the (weighted) average order of the Δ -function (exponential-sum / Dirichlet-series saddle-point analysis; not a critical exponent in the renormalization-group sense).

For the present paper, the only fact about K_{AUD} we use is that it is a specific real constant with the numerical value above. The bound $\eta \leq 2G$ is reported as an observational claim about that numerical value.

4. The Wilson-Fisher family scope criterion

The bound's scope is the set of $d = 3$ critical points satisfying all four of the following criteria:

1. Unitary CFT. The critical theory is a unitary conformal field theory at the critical point. Anomalous dimensions are real and positive. This excludes non-unitary critical points such as percolation (which has $\eta = -0.0431$ by analytic continuation of an underlying non-unitary CFT) and the Lee-Yang edge singularity.
2. Positive correlation-length exponent ν . The correlation length diverges at the critical point: $\nu > 0$. This is satisfied at standard second-order transitions and excludes pathologies of correlated-disorder or $T = 0$ fixed points where ν is not the standard quantity.
3. Standard hyperscaling with two independent exponents. The critical exponents satisfy $2 - \alpha = \nu \cdot d$ with the standard Rushbrooke/Fisher/Widom relations, and the universality class is specified by two independent exponents η and ν (all other exponents are derived). This excludes $T = 0$ disorder fixed points where modified hyperscaling $2 - \alpha = \nu \cdot (d - \theta)$ introduces a third independent exponent θ (e.g., random-field Ising).
4. Finite-temperature continuous transition. The transition occurs at finite $T > 0$ and is genuinely second-order (not weakly first-order, pseudo-critical, or with runaway flow). This excludes DQCP-SU(2)-type pseudo-critical transitions and disorder-induced-second-order transitions whose underlying pure transition is first-order.

These four criteria collectively pick out the ϕ^4 -like universality classes whose physics is the dimensional continuation of the Wilson-Fisher fixed point from $d = 4 - \epsilon$. They are physics-tier criteria, established independently of any framework consideration; we adopt them as the scope of the bound because (a) they are the natural domain of the WF fixed-point family in standard critical-phenomena theory, and (b) the bound holds within this domain across every class tested while failing in classes that violate any one of the four criteria.

4.1 Justification of the four criteria as physics-tier (not framework-tier)

Each of the four criteria has independent physical motivation:

- Unitary CFT: required for a probability interpretation; non-unitary critical points (percolation, Lee-Yang) have negative or complex η , which makes the " $\eta \leq 2G$ " comparison structurally different.
- Positive ν : required for the correlation length to diverge in the standard sense; $T = 0$ fixed points and certain disorder fixed points have qualitatively different ν -behavior.
- Standard hyperscaling: the bound is on $(2 - \eta)$, the effective correlation dimension;

modified hyperscaling at $T = 0$ fixed points (RFIM) introduces a third exponent θ , changing the structural meaning of $(2 - \eta)$.

- Finite-T second order: pseudo-criticality (DQCP-SU(2)) and disorder-induced second-order behavior of an underlying first-order transition (disordered Potts large- q) are increasingly understood not to be genuine continuous CFTs in the unitary sense.

The bound's scope is therefore narrowed by independent physics, not by ad-hoc framework choice. A class that fails any one of the four criteria is outside the bound's claim because its critical-phenomena structure is structurally different from the WF family, not because the framework wishes to exempt it.

5. Empirical universality survey

We test the bound $\eta \leq 2G$ against ten WF-family classes at $d = 3$, ranging from conformal-bootstrap precisions of $\sigma_\eta \sim 2 \times 10^{-6}$ (3D Ising, Kos et al. 2016) and propagated-from-pivot-Monte-Carlo $\sigma_\eta \sim 3 \times 10^{-6}$ (SAW, derived from Clisby-Dünweg 2016 ν and Clisby 2017 γ via Fisher's relation) down to 4-loop perturbative RG and current Monte Carlo $\sigma_\eta \sim 2 \times 10^{-3}$ (random-bond Ising and the MN cubic classes). The table below summarizes the test; citations follow the table.

5.1 The ten WF-family classes

Class	$\eta \pm \sigma$	$2G - \eta$	σ -margin	Source
SAW (0(0))	0.0310434 ± 0.0000030	$+0.008440$	$+2813\sigma$	Clisby 2016 + 2017 ($\eta = 2 - \gamma/\nu$)
3D Ising (0(1))	0.0362978 ± 0.0000020	$+0.003186$	$+1593\sigma$	Kos et al. 2016 (CB)
3D XY (0(2))	0.038176 ± 0.000044	$+0.001308$	$+29.7\sigma$	Chester et al. 2020 (CB)
3D Heisenberg (0(3))	0.03810 ± 0.00020	$+0.001384$	$+6.9\sigma$	Kos et al. 2016 (CB)
3D O(4)	0.0365 ± 0.0010	$+0.002984$	$+3.0\sigma$	Monte Carlo
Cubic anisotropy $N = 3$	$\approx \eta(0(3))$	$\approx +0.0014$	$\approx +7\sigma$	Hasenbusch 2022
Cubic anisotropy $N = 4$	0.0371 ± 0.0002	$+0.002384$	$+11.9\sigma$	Hasenbusch 2022
MN cubic $N = 2$	0.0343 ± 0.0020	$+0.005184$	$+2.6\sigma$	Mudrov-Varnashev 2001
MN cubic $N = 3$	0.0345 ± 0.0015	$+0.004984$	$+3.3\sigma$	Mudrov-Varnashev 2001
Random-bond Ising (randomly dilute)	0.035 ± 0.002	$+0.004484$	$+2.2\sigma$	CMM-PV 2003; Hasenbusch et al. 2007

All ten classes satisfy $\eta < 2G$. No class measured to date violates the bound. The σ -margins span four orders of magnitude, from the $+2.2\sigma$ tightness at randomly dilute Ising (the priority falsification-risk target identified in §7) to the $+2813\sigma$ slack at SAW (derived from Clisby-Dünweg 2016 and Clisby 2017 pivot Monte Carlo via Fisher's relation $\eta = 2 - \gamma/\nu$) and the $+1593\sigma$ slack at 3D Ising (Kos et al. 2016 conformal bootstrap), where the bound is far from the data even at the highest currently-available precision.

Note on the SAW derivation chain. The 3D SAW η value used in the table above is derived from two independent high-precision measurements via Fisher's relation $\eta = 2 - \gamma/\nu$: $\nu = 0.58759700(40)$ from Clisby-Dünweg 2016 (pivot-algorithm Monte Carlo on chains up to $N \approx 3.4 \times 10^7$ monomers) and $\gamma = 1.15695300(95)$ from Clisby 2017 (scale-free pivot Monte Carlo). The derivation yields $\eta = 0.0310434(21)$ with σ_η propagated from σ_γ and σ_ν . The $\sigma_\eta = 0.0000030$ quoted in the table above is mildly conservative relative to rigorous propagation ($\sigma_\eta \approx 2.1 \times 10^{-6}$ via standard error propagation, a factor $1.4\times$ tighter than the conservative value used); the $+2813\sigma$ margin is correspondingly conservative.

5.2 The tightest unitary WF-family case: random-bond Ising – worked example

Because random-bond Ising is the priority falsification-risk target, we walk through the case in full so a skeptical reader can evaluate the bound's empirical position without further reference work.

(a) Definition. The randomly dilute (random-bond) Ising model in $d = 3$ is the standard Ising Hamiltonian $H = -\sum J_{ij} s_i s_j$ with quenched random bonds J_{ij} . Two equivalent realizations are studied: site-dilution (random fraction of spins replaced by vacancies) and bond-dilution (random fraction of bonds with $J_{ij} = 0$). Both flow to the same fixed point in $d = 3$, characterized by the disordered Ising universality class, distinct from both pure Ising and random-field Ising. The transition is at finite temperature $T_c > 0$ and is genuinely second-order under quenched disorder.

(b) WF-family scope verification.

1. Unitary CFT: The disordered Ising fixed point in $d = 3$ is a unitary CFT (anomalous dimensions are real and positive; the operator content is well-defined). Confirmed by perturbative-RG convergence in the literature and by Monte Carlo agreement (Calabrese-Martin-Mayor-Pelissetto-Vicari 2003) with the bootstrap-style expectations.
2. Positive ν : $\nu = 0.683(2)$ (Hasenbusch et al. 2007), well-defined and positive.
3. Standard hyperscaling: The class satisfies standard hyperscaling $2 - \alpha = \nu \cdot d$ with two independent exponents. There is no third exponent θ (in contrast with RFIM); the universality class is fully characterized by (η, ν) . This is the criterion that distinguishes randomly dilute Ising from RFIM, even though both involve disordered Ising spins.
4. Finite-T second-order transition: $T_c > 0$; transition is genuinely second-order under quenched bond/site disorder.

All four WF-family criteria are satisfied. The class is squarely within the bound's scope.

(c) Measurement source. The current best η for $d = 3$ randomly dilute Ising:

- Hasenbusch, Parisen Toldin, Pelissetto, and Vicari, Universality class of three-dimensional site-diluted and bond-diluted Ising systems, arXiv:cond-mat/0611707 (2007): high-precision Monte Carlo simulations, finding $\eta \approx 0.036(1)$ for site-dilution and consistent values for bond-dilution.
- Calabrese, Martin-Mayor, Pelissetto, and Vicari, The three-dimensional randomly dilute Ising model: Monte Carlo results, Phys. Rev. E 68, 036136 (2003), arXiv:cond-mat/0306272: high-statistics Monte Carlo on cubic lattices L^3 with $L \leq 256$ at density $x = 0.8$, giving $\eta = 0.035(2)$, $\nu = 0.683(3)$.

The consensus value, weighted by the methods' mutual consistency, is $\eta = 0.035 \pm 0.002$ – a slight conservative widening of the Hasenbusch et al. σ to accommodate the Monte Carlo spread between the two independent studies (Calabrese et al. 2003 at $\eta = 0.035(2)$; Hasenbusch et al. 2007 at $\eta = 0.036(1)$).

(d) Numerical comparison to 2G.

- $\eta = 0.035 \pm 0.002$
- $2G = 0.039484$
- $2G - \eta = 0.004484$
- $\sigma\text{-margin} = 0.004484 / 0.002 = +2.24\sigma$

The bound $\eta \leq 2G$ is satisfied at $+2.2\sigma$ (or $+2.24\sigma$ to two decimals). This is the smallest σ -margin among all ten WF-family classes tested.

(e) What a higher-precision measurement would test.

A factor-of-three precision improvement ($\sigma_\eta \approx 0.0007$, achievable with a next-generation Monte Carlo simulation at improved lattice sizes and statistics) would have two possible outcomes:

- If the central η stays near 0.035: the σ -margin becomes $(0.004484 / 0.0007) \approx +6.4\sigma$. The bound becomes well-supported at the random-bond Ising case.
- If the central η drifts upward toward 2G: a central value of, say, $\eta = 0.038$ with $\sigma_\eta = 0.0007$ would place 2G at $+2.1\sigma$ above η – still satisfied, but the $+6\sigma$ margin is no longer there.
- If the central η crosses 2G: $\eta \geq 0.039484$ with $\sigma_\eta = 0.0007$ would be a 3σ -or-better violation of the bound, satisfying the falsification commitment of §7.5.

The random-bond Ising case is therefore the single sharpest experimental test of the framework's empirical prediction available with foreseeable technology.

5.3 The 3D Ising case: 1593 σ below 2G

The 3D Ising universality class is the tightest η measurement in the entire critical-phenomena literature, with conformal-bootstrap precision $\sigma_\eta \approx 2 \times 10^{-6}$ (Kos et al. 2016). At this precision, the bound $\eta \leq 2G$ is satisfied at 1593 σ – meaning the bound has 1593 standard deviations of margin in the most precisely measured WF-family case. This makes 3D Ising the least useful case for falsifying the bound and the most useful case for confirming that the bound is not at the edge of measurement.

5.4 The σ -margin ranking

Rank	Class	σ -margin to 2G
1	Random-bond Ising	+2.2 σ
2	MN cubic $N = 2$	+2.6 σ
3	O(4)	+3.0 σ
4	MN cubic $N = 3$	+3.3 σ
5	Heisenberg (O(3))	+6.9 σ
6	Cubic anisotropy $N = 3$	$\approx +7\sigma$
7	Cubic anisotropy $N = 4$	+11.9 σ
8	XY (O(2))	+29.7 σ
9	3D Ising	+1593 σ
10	SAW (O(0))	+2813 σ

The top of the ranking (random-bond Ising at $+2.2\sigma$) is where future precision improvements would most quickly test the bound. The bottom (SAW at $+2813\sigma$ and 3D Ising at $+1593\sigma$) establishes that the bound is genuinely above the data at the highest precisions currently available – far above, not a borderline match. SAW's apparent looseness in earlier framework-internal discussion was an artifact of the older $\sigma \approx 0.003$ Monte Carlo/consensus precision; with Clisby-Dünweg 2016's $\nu = 0.58759700(40)$ and Clisby 2017's $\gamma = 1.15695300(95)$ propagated via Fisher's relation, $\sigma_\eta \approx 3 \times 10^{-6}$ (mildly conservative relative to the $\approx 2.1 \times 10^{-6}$ from rigorous propagation), and SAW becomes one of the two highest-precision bounds-confirming cases.

5.5 Null returns honestly recorded

The following WF-family-relevant classes lack rigorous η values at the precision needed for this test, and we record them as open empirical-measurement targets rather than as evidence:

- 3D lattice gauge theories with clean global-symmetry continuous transitions (no rigorous η values located at comparable precision).
- O(N) with $N \in \{5, 6, 7, \dots\}$ ($1/N$ expansion estimates exist but at insufficient precision

- for sub- σ testing of $\eta = 2G$).
- Cubic anisotropy at $N \geq 5$ (no high-precision data located).
- MN cubic at $N \geq 4$ (no high-precision data located).
- $CP^*(N-1)$ models at $d = 3$ large N (no η at comparable precision).
- WZW / σ -model fixed points at $d = 3$ (no η at sub-percent precision).

These remain open. The bound is not claimed to be tested against these classes; its scope of test is the ten classes above.

6. Classes outside WF-family scope

Three classes are known to have $\eta > 2G$ at $d = 3$ and would, if treated naively as counter-examples, falsify the bound. Each is outside the WF-family scope for independent physics reasons – reasons that predate and are independent of the framework, and that would be the right scope-boundary even in the absence of any framework consideration.

6.1 Random-Field Ising Model (RFIM) at $d = 3$

Quantity	Value	Bound comparison
η (Monte Carlo, Rieger 1995 and follow-ups)	≈ 0.5	12× above $2G$
η (experiment, Belanger and collaborators)	≈ 0.16	4× above $2G$

Exclusion reason (criterion 3 violated): RFIM is governed by a $T = 0$ fixed point with strong quenched disorder. The transition obeys modified hyperscaling $2 - \alpha = \nu \cdot (d - \theta)$, where θ is a third independent exponent characterizing the dimensional reduction $d \rightarrow d - \theta$ associated with supersymmetry restoration in the dimensional-reduction picture (Fytas–Martín-Mayor 2017, 2019). The presence of θ means RFIM critical phenomena are specified by three independent exponents (η, ν, θ), not two. The structural meaning of $(2 - \eta)$ at RFIM is therefore different from its meaning at WF-family critical points; the bound's framing in terms of $(2 - \eta)$ is not the right framing for RFIM.

RFIM is not a counter-example to the bound; it is outside the bound's scope.

6.2 Deconfined SU(2) Quantum Critical Point (J-Q model)

Quantity	Value	Bound comparison
η (Sandvik 2007, J-Q model)	≈ 0.26	6.6× above $2G$

Exclusion reason (criterion 4 violated): Recent consensus (Nahum et al. 2015; Wang et al. 2017; Song et al. 2025, arXiv:2307.02547) increasingly favors the interpretation that the SU(2) deconfined critical point is pseudo-critical or weakly first-order, with runaway RG flow and a complex CFT in its description. Under this interpretation it is not a genuine continuous CFT in the unitary sense; the measured " η " is an effective scaling exponent that diverges in the thermodynamic limit but does not correspond to the operator dimension of a unitary CFT at the transition.

DQCP-SU(2) is not a counter-example to the bound; it is outside the bound's scope because criterion 4 (genuine continuous CFT) is violated.

6.3 Disordered Potts at large q

Quantity	Value	Bound comparison
η (derived from $2\beta/\nu - 1$, Chatelain et al.)	≈ 0.20	5× above $2G$

Exclusion reason (criterion 4 violated): Pure Potts at large q in $d = 3$ is first-order. Quenched bond disorder rounds the transition to second-order (Imry–Wortis / Aizenman–Wehr theorem). The resulting "second-order" critical point is disorder-induced and the validity of standard hyperscaling at such disorder-induced-second-order points is debated. The transition is not a ϕ^4 -type WF fixed-point critical point in the standard sense.

A second-tier caveat on the η value itself. The $\eta \approx 0.20$ figure above is not a direct measurement; it is inferred via the hyperscaling-style relation $\eta = 2\beta/\nu - 1$ from a measured $\beta/\nu \approx 0.60$. The inference assumes standard hyperscaling – exactly the assumption that criterion 3 of WF-family scope places under question for disordered-second-order points. So even if the inferred $\eta \approx 0.20$ were a valid characterization of the class, the class cannot function as a true counter-example to the bound: either the hyperscaling assumption holds (and the class fails criterion 4 because the transition is disorder-induced from a first-order pure transition) or hyperscaling fails (and the η inference itself is invalid). Both readings place the class outside the bound's scope without making it a falsifying case.

Disordered Potts large- q is not a counter-example to the bound; it is outside the bound's scope.

6.4 The structural symmetry of the survey

The bound's scope is sharpened, not weakened, by the exclusion of these three classes. The exclusion is physics-tier ($T = 0$ fixed point, pseudo-criticality, disorder-induced second order) and would be the right scope-boundary even if the framework did not exist. The combination is striking:

- Every class satisfying the four WF-family criteria has $\eta < 2G$ (ten cases, range from +2.2 σ at randomly dilute Ising to +2813 σ at SAW).
- Every class with $\eta > 2G$ that we have located in the literature violates at least one of the four criteria.

This is the empirical heart of the paper: the bound and the scope criterion are mutually consistent across the published data.

7. Predictions and falsifiability

The bound makes the following predictions that can be tested by future measurements:

7.1 Priority falsification target: random-bond Ising at improved precision

A higher-precision determination of η for the $d = 3$ randomly dilute (random-bond) Ising universality class – by either:

- A conformal-bootstrap calculation reaching $\sigma_\eta \approx 0.0007$ (factor-3 improvement over current Monte Carlo),
- A large-scale Monte Carlo simulation at improved system sizes and statistics,

would test the bound at the +6 σ level (if $\eta \approx 0.035$ holds) or, alternatively, place $\eta = 2G$ at the edge of measurement (if η drifts upward toward $2G$ with precision). Either outcome is informative. This is the single cleanest empirical falsification route the bound has.

7.2 Secondary targets within WF-family scope

The next-tightest cases – MN cubic $N = 2$ at +2.6 σ , $O(4)$ at +3.0 σ , MN cubic $N = 3$ at +3.3 σ – are also useful falsification routes. Tightening any of them by a factor of 3–5 in precision would either confirm $\eta < 2G$ at the >10 σ level or push $\eta = 2G$ into the measurement range.

7.3 New universality classes within WF-family scope

Any new $d = 3$ critical point identified as a unitary CFT with positive ν , standard hyperscaling, and finite-T continuous transition is a test of the bound. Specific candidates with no current high-precision η :

- 3D lattice gauge theories with clean global-symmetry continuous transitions.
- Cubic anisotropy at $N \geq 5$.
- MN cubic at $N \geq 4$.
- $CP^{(N-1)}$ models at $d = 3$ large N .

A measurement of η at any of these to sub-percent precision is an unambiguous test.

7.4 The conformal-bootstrap derivation route

A direct first-principles derivation of $\eta \leq 2G$ from the conformal-bootstrap machinery (modular bootstrap, dispersion relations, sum rules at the Wilson-Fisher fixed point) would promote the bound from observation to theorem and would resolve the structural question of why K_{AUD} specifically is the right ceiling. Pelissetto–Vicari (arXiv:2510.17637, 2025/rev 2026) conjecture a related dimensional-operator-inequality lower bound $\nu \geq 1/(2 - \eta)$ at the Wilson-Fisher fixed point; combining the present observation $\eta \leq 2G$ with Pelissetto–Vicari's conjectural $\nu \geq 1/(2 - \eta)$ gives the derived lower bound

$$\nu \geq 1/(2K_{\text{AUD}}) \approx 0.510$$

across WF-family classes (a derivation in this paper from the two independent inputs; the combined result inherits the conjectural status of its inputs and is not stated in either source paper). The Pelissetto–Vicari route and the corridor framing are independent attempts at related questions; their compatibility lends modest evidence to both.

7.5 What a single counter-example would do

A single $d = 3$ unitary CFT with positive ν , standard hyperscaling, and finite-T continuous transition having measured $\eta > 2G$ at >3 σ confidence would falsify the bound as stated. We commit, in advance, to accepting such a measurement as falsification. The bound is not framework-protected: it is an observational claim with a definite scope, and it is falsifiable by a measurement that satisfies all four scope criteria and exceeds $2G$.

8. Why specifically $2G$?

This section is brief by design: the derivation of why K_{AUD} (and therefore $2G$ via Fisher's relation) might be the relevant constant is the open theoretical problem and belongs in a different paper. Here we give only the minimum needed for a reader to understand the observation's structure.

8.1 Fisher's relation supplies the factor of 2

The "2" in " $2G$ " is not a framework-applied doubling. It is the upper endpoint of the $(2 - \eta)$

coordinate, supplied by Fisher's exact relation $\gamma = v(2 - \eta)$ and the dimensional sum at the upper critical dimension. At the Gaussian fixed point (mean field, $\eta = 0$), $(2 - \eta) = 2$ exactly. This is established RG theory.

The framework's claim is about the lower endpoint of $(2 - \eta)$. The framework conjectures that $K_{\text{AUD}} = \sqrt{2} \cdot \ln 2$ is the structurally meaningful lower bound on $(2 - \eta)$ at the Wilson-Fisher fixed point. The corridor width $2 - 2K_{\text{AUD}}$ then simplifies to $2G$ algebraically, using $K_{\text{AUD}} + G = 1$.

Endpoint	Value	Source
Upper (mean field, $\eta = 0$)	$(2 - \eta) = 2$	Fisher's relation
Lower (anomalous, $\eta = 2G$)	$(2 - \eta) = 2K_{\text{AUD}}$	Framework conjecture
Width	$2 - 2K_{\text{AUD}} = 2G$	Algebra

Pushback on the factor of 2 is misplaced – that is Fisher's. The right target for pushback is the framework's lower-bound conjecture, which is the open Tier 6 derivation question addressed in Part II §11.5 (the named Tier 6 derivation targets) and §15.2 (the framework-architecture open questions).

8.2 Why this value and not a tighter one?

A natural objection is "why $2G$ rather than a tighter ceiling that saturates the data more closely?" We tested nearby candidate ceilings against the tightest WF-family case (randomly dilute Ising at $\eta = 0.035 \pm 0.002$) and found that the framework's $2G$ value is empirically robust:

Candidate ceiling	Value	σ -margin at random-bond Ising	Falsification risk under improved precision
$2G = 2(1 - K_{\text{AUD}})$ (framework)	0.039484	+2.24 σ	Lowest of the four candidates
$\pi/80$	0.039270	+2.13 σ	Slightly higher; still well above data
$1/25.5$	0.039216	+2.11 σ	Slightly higher; still well above data
$2(1 - \ln(8/3))$	0.038341	+1.67 σ	Highest; $\sim \times 2$ precision $\rightarrow$ within 1σ

(Margins computed as $(\text{ceiling} - \eta) / \sigma_\eta$ at $\eta = 0.035$, $\sigma_\eta = 0.002$. All four candidates currently satisfy $\eta < \text{ceiling}$. The tighter ceilings are not violated by current data but sit closer to the data, so they would be the first to fail if the central η drifts upward or if measurement precision improves.)

The differences between candidate ceilings at random-bond Ising are small in absolute terms (0.04–0.13 in σ -margin), so $2G$'s empirical advantage over nearby alternatives is not large at the current precision. What it does establish is that $2G$ is the most robust of the four candidates against foreseeable precision improvements. At the CFT-bootstrap cases (Ising +1593 σ , XY +29.7 σ , Heisenberg +6.9 σ) the difference between candidates is negligible – all are far above data – so the empirical-robustness argument is driven by the tightest cases (random-bond Ising and, secondarily, MN cubic $N=2$ and SAW).

This is not an argument that $2G$ is unique – the framework's claim is that K_{AUD} specifically (rather than nearby ceilings) is the right one for structural reasons developed in Part II below. The robustness argument is a complementary empirical observation: among nearby candidates, the framework's choice happens to be the value most likely to survive next-generation precision improvements at the tightest current cases.

Part II – The Mathematical Architecture

Part I reports the empirical observation $\eta \leq 2G$ across ten Wilson-Fisher-family universality classes in $d = 3$, deferring the question of why this particular value of the ceiling. Part II takes up that question.

The path of the argument is the following. §9 studies the algebraic structure of the dome function $g(x) = \tanh(x)/\cosh(2x)$, known from hyperbolic-3-manifold geometry as the Hodgson-Kerckhoff tube-packing function. §9 opens with a master landmark table organizing the dome's five algebraically distinguished landmarks (origin $u = 1$, K_{AUD} -defining position $u = 5/4$, self-reference $u = \sqrt{2}$, peak $u = \phi$, binary-doubling $u = 2$) and the framework findings that sit on each; the subsequent subsections prove the structural properties the table claims, including the result that the dome's own internal algebra independently distinguishes both ingredients of $K_{\text{AUD}} = \sqrt{2} \cdot \ln(2)$ and that the two HK landmarks ($u = 5/4$ and $u = 2$) sit in distinct exact algebraic environments ($\sqrt{2}$ -rational and $\sqrt{3}$ -rational, respectively). In §10 we recall that this same dome function appears, with no free parameters, as the resummation factor $R(N)$ for the anomalous dimension at the 3D Wilson-Fisher fixed point across $N = 0-4$. In §11 we state the Central Synchronization Conjecture: that K_{AUD} 's recurrence across multiple independent derivation chains (H_4 polytope geometry, Algorithm EA saturation at the binary cell, Hodgson-Kerckhoff tube-packing, the Wilson-Fisher ceiling) is the same geometry-information intersection at $n = 2$ showing up in different mathematical clothing. In §12 we report a sharper structural identity at the upper critical dimension $d_c = 4$ where multiple framework-relevant identities align. In §13 we close with the dimensional context – why H_4 and $d_c = 4$ are the same number from independent sources, and why the corridor is sharply dimension-specific.

Throughout Part II we maintain the tier distinction of Part I: theorems are exact algebraic or numerical identities verified at high precision; observations are empirical or structural matches lacking a derivation; conjectures are proposed unifying mechanisms not yet proved. The

Central Synchronization Conjecture itself stands as a conjecture; its supporting items are individually at Theorem-tier or Observation-tier with high-precision verification.

9. The dome's algebraic architecture

This section is structurally a tour of the dome's algebraic landmarks. Each subsection unpacks one landmark or one structural property; the master table below organizes them at a glance, with the theorems that follow being rigorous statements of what the table claims.

Master landmark table. In $u = \cosh(2x)$ coordinates, the dome function $g(u) = \sqrt{(u-1)/(u+1)}$ has five algebraically distinguished landmarks across $u \in [1, \infty)$:

Landmark $u = 1$ – dome origin (boundary)

- $x = \operatorname{arccosh}(u)/2 = 0$
- $g(u) = 0$
- Selected by: $u = 1$ (boundary)

Landmark $u = 5/4$ – HK K_{AUD} -defining position

- $x = (\ln 2)/2$
- $g(u) = 4/15$ (exact)
- Selected by: $x = (\ln 2)/2 = \operatorname{arcsinh}(1/(2\sqrt{2}))$
- Identification: HK Theorem 4.4; $c = 2 \ln 2$ lands dome here at $d_c = 4$ (§12)

Landmark $u = \sqrt{2}$ – framework's geometric primitive

- $x = \operatorname{arccosh}(\sqrt{2})/2$
- $g(u) = 1 - 1/\sqrt{2}$
- Selected by: $g'/g = g$, unique self-reference (Theorem 9.4)
- Identification: framework's geometric primitive associated with H_4 (Paper 3 V3 §3.4 catalogues four origins for $\sqrt{2}$ in H_4 geometry; see §13.3 convention note for the relationship to standard unit-edge polytope circumradius)

Landmark $u = \phi$ – dome peak (golden ratio)

- $x = \operatorname{arccosh}(\phi)/2$
- $g(u) = \phi^{(-5/2)}$
- Selected by: $P(u) = u^2 - u - 1 = 0$ (Cor 9.3)
- Identification: dome peak; framework's geometric primitive associated with H_3 (overflow case in §13.3 within the framework's per- H_n primitive convention)

Landmark $u = 2$ – HK critical tube radius (binary doubling)

- $x = R_0 = \operatorname{arccosh}(2)/2$
- $g(u) = 1/(2\sqrt{3})$
- Selected by: $\cosh(2x) = 2$ (binary doubling)
- Identification: $h(R_0) = K_{\text{AUD}}$ (Theorem 10.1); triple convergence point (§10.8)

The dome's geography is the sequence $1 \rightarrow 5/4 \rightarrow \sqrt{2} \rightarrow \phi \rightarrow 2$: from the origin, through the K_{AUD} -defining position, through the geometric primitive, through the peak, to the binary-doubling boundary. Every algebraic structure unpacked in the remainder of §9 – the factorization, the golden polynomial, the self-reference, the integral, the dilation-operator structure, the K_{AUD} -ingredient generation – is a property of this geography.

Two further structural facts about this geography are worth stating up front, because they organize what follows:

1. The framework's two primary constants are generated by the dome's two internal structures. The geometric primitive $\sqrt{2}$ emerges from the self-reference landmark (Theorem 9.4); the informational primitive $\ln 2$ emerges as twice the dome's integral mass (Theorem 9.5). Their product $K_{\text{AUD}} = \sqrt{2} \cdot \ln 2$ enters the framework as a multiplicative combination of the dome's two distinguished landmarks, not as an externally chosen constant.
2. The two HK landmarks ($u = 5/4$ and $u = 2$) sit in distinct exact algebraic environments. At Landmark ($5/4$), all six hyperbolic values at $x = (\ln 2)/2$ are rational or $\sqrt{2}$ -rational: $\sinh(x) = 1/(2\sqrt{2})$, $\cosh(x) = 3/(2\sqrt{2})$, $\tanh(x) = 1/3$, $\sinh(2x) = 3/4$, $\cosh(2x) = 5/4$, $\tanh(2x) = 3/5$. At Landmark (2), all six hyperbolic values at $x = R_0$ are rational-in- $\sqrt{3}$: $\sinh(R_0) = 1/\sqrt{2}$, $\cosh(R_0) = \sqrt{3}/2$, $\tanh(R_0) = 1/\sqrt{3}$, $\sinh(2R_0) = \sqrt{3}$, $\cosh(2R_0) = 2$, $\tanh(2R_0) = \sqrt{3}/2$. The dome thus has two closed algebraic worlds at its two HK landmarks, one built on $\sqrt{2}$ and one built on $\sqrt{3}$. See §9.10 below.

With the master table in place, the remaining subsections of §9 proceed as proofs and structural unpacking of what the table claims.

9.1 The dome function and two coordinate representations

We define the dome function by

$$g(x) = \tanh(x) / \cosh(2x), \quad x \geq 0.$$

It is positive on $x > 0$, vanishes at $x = 0$ and as $x \rightarrow \infty$, and is smooth with a single interior

maximum. In the substitution $u = \cosh(2x)$ (so that $u \geq 1$, with $u = 1$ corresponding to $x = 0$ and $u \rightarrow \infty$ to $x \rightarrow \infty$), the dome admits the equivalent algebraic representation

$$g(u) = \sqrt{u - 1} / [u \cdot \sqrt{u + 1}].$$

The two representations are related by $\tanh^2(x) = (u - 1)/(u + 1)$ and the identity $\cosh(2x) = u$. The u -representation makes the function's algebraic landmarks transparent; the x -representation makes its spectral and integral structure transparent. We move between the two as convenience dictates.

9.2 The sech^2 / doubling-rate factorization

Theorem 9.1. The dome function admits the factorization

$$g(x) = (1/2) \cdot \text{sech}^2(x) \cdot \tanh(2x).$$

Proof. Using $\sinh(2x) = 2 \sinh(x) \cosh(x)$,

$$(1/2) \cdot \text{sech}^2(x) \cdot \tanh(2x) = (1/2) \cdot (1/\cosh^2(x)) \cdot \sinh(2x)/\cosh(2x) = (1/2) \cdot (1/\cosh^2(x)) \cdot 2 \sinh(x) \cosh(x) / \cosh(2x) = \sinh(x) / [\cosh(x) \cdot \cosh(2x)] = \tanh(x) / \cosh(2x) = g(x). \blacksquare$$

The factorization separates the dome into two physically distinct components. The factor $\text{sech}^2(x)$ is the universal spectral kernel for damped harmonic oscillators near resonance – Lorentzian in frequency, of width set by the damping rate. The factor $\tanh(2x)$ is the binary-doubling carrier: it is the saturation function with characteristic scale set by the doubled coordinate $2x$, equivalently the operator that converts an additive variable into a saturating "binary distinction." The dome is the product of these two: a spectral kernel modulated by binary-doubling saturation.

The structural reading of Theorem 9.1 is that the doubling rate is not imposed on the dome from outside (via Fisher's relation $\gamma = v(2 - \eta)$, via the framework's binary architecture, via any external narrative). It is part of the dome's intrinsic algebraic identity. Both sech^2 and $\tanh(2x)$ appear as factors in the function itself; neither is added.

Numerical verification of Theorem 9.1 at $\text{dps} = 80$ across the test grid $x \in \{0.1, 0.3, 0.5, 0.7, 1.0, 1.5, 2.0, 3.0\}$: residuals are zero or below 10^{-82} (machine precision at the test dps). The identity is exact.

9.3 The logarithmic derivative and the golden polynomial

Theorem 9.2. In $u = \cosh(2x)$ coordinates, the dome function has logarithmic derivative

$$d(\ln g)/du = -(u^2 - u - 1) / [u(u^2 - 1)].$$

Proof. Differentiate $\ln g(u) = (1/2) \ln(u - 1) - \ln u - (1/2) \ln(u + 1)$ and combine fractions. The numerator that appears after combination is exactly $-(u^2 - u - 1)$; the denominator is $u(u^2 - 1)$. $\blacksquare$

The numerator $P(u) = u^2 - u - 1$ is the golden polynomial. Its positive real root is $\phi = (1 + \sqrt{5})/2$, the golden ratio.

Corollary 9.3. The dome's peak is at $u = \phi$, with critical value $g(\phi) = \phi^{-5/2}$.

Proof. $g'(u) = 0$ requires the numerator of $d(\ln g)/du$ to vanish, which gives $P(u) = 0$, hence $u = \phi$. Direct computation yields $g(\phi) = \phi^{-5/2}$. $\blacksquare$

The peak of the dome is therefore at the golden ratio – exactly, not approximately. The golden polynomial enters not as a numerical match but as the algebraic structure governing the dome's variation in u -coordinates.

9.4 Three algebraically distinguished landmarks

The polynomial $P(u) = u^2 - u - 1$ takes three distinguished values within the dome's u -range:

Landmark	u	$P(u)$	$g(u)$	Selection condition
Peak	ϕ	0	$\phi^{-5/2}$	$P(u) = 0$, gives $g'(u) = 0$
Self-reference	$\sqrt{2}$	$1 - \sqrt{2}$	$1 - 1/\sqrt{2}$	$g'/g = g$ (Theorem 9.4 below)
Unit polynomial	2	1	$1/(2\sqrt{3})$	$P(u) = 1$ (binary-doubling argument)

The three landmarks are selected by three different algebraic conditions – vanishing of P , the self-reference equation, and P attaining unity. The framework's three primary constants $\{\phi, \sqrt{2}, 2\}$ appear at these three landmarks; they are not imported from outside but selected by the dome's own algebra.

9.5 The self-reference at $\sqrt{2}$

Theorem 9.4. The condition $g'(u)/g(u) = g(u)$ – that the logarithmic derivative of the dome equals the dome's value – has a unique positive solution: $u = \sqrt{2}$.

Proof. From Theorem 9.2, $g'(u)/g(u) = -P(u) / [u(u^2 - 1)]$ where $P(u) = u^2 - u - 1$. From §9.1, $g(u) = \sqrt{u - 1} / [u \cdot \sqrt{u + 1}]$. The equation $g'/g = g$ becomes

$$-(u^2 - u - 1) / [u(u^2 - 1)] = \sqrt{u - 1} / [u \cdot \sqrt{u + 1}].$$

Cancel u from both sides and use $u^2 - 1 = (u - 1)(u + 1)$:

$$-(u^2 - u - 1) / [(u - 1)(u + 1)] = \sqrt{u - 1} / \sqrt{u + 1}.$$

Multiply both sides by $(u - 1)(u + 1)$ and simplify the right side using $(u - 1)(u + 1)/\sqrt{u + 1} = (u - 1)\sqrt{u + 1}$:

$$-(u^2 - u - 1) = (u - 1)^{3/2} \cdot (u + 1)^{1/2} = \sqrt{[(u - 1)^3(u + 1)]}.$$

The left side is positive precisely when $u^2 - u - 1 < 0$, equivalently $u < \phi$. In that range both sides are positive, and squaring gives

$$(u^2 - u - 1)^2 = (u - 1)^3(u + 1).$$

Expanding: the left side is $u^4 - 2u^3 - u^2 + 2u + 1$; the right side is $u^4 - 2u^3 + 2u - 1$. Their difference is

$$(u^4 - 2u^3 - u^2 + 2u + 1) - (u^4 - 2u^3 + 2u - 1) = -u^2 + 2.$$

Setting this to zero gives $u^2 = 2$, with the unique positive solution $u = \sqrt{2}$. Since $\sqrt{2} \approx 1.414 < \phi \approx 1.618$, this solution lies in the regime where the squaring step is valid. ■

At $u = \sqrt{2}$: $g(\sqrt{2}) = (\sqrt{2} - 1)/\sqrt{2} = 1 - 1/\sqrt{2} \approx 0.2929$, and indeed $g'(\sqrt{2})/g(\sqrt{2}) = 1 - 1/\sqrt{2} = g(\sqrt{2})$.

The geometric content of Theorem 9.4 is the following. The H_4 regular polytope (the 600-cell, dual of the 120-cell) has circumradius $\sqrt{2}$, and $\sqrt{2}$ appears as the framework's geometric primitive by polytope geometry. Theorem 9.4 shows that the dome function – defined without any reference to H_4 or to the framework – independently distinguishes the same value $\sqrt{2}$ as its unique self-referential fixed point. The defining equation of $\sqrt{2}$ ($u^2 = 2$) emerges from the dome's own dynamics.

9.6 The dome integral identity

Theorem 9.5. The dome function satisfies

$$\int_0^\infty g(x) dx = \int_0^\infty \tanh(x)/\cosh(2x) dx = \ln(2) / 2.$$

Proof. Substitute $t = \cosh(x)$, so $dt = \sinh(x) dx$ and $\cosh(2x) = 2t^2 - 1$. The integrand becomes $\tanh(x)/\cosh(2x) \cdot dx = \sinh(x)/[\cosh(x) \cdot \cosh(2x)] \cdot dx = dt/[t(2t^2 - 1)]$. Partial fractions:

$$1/[t(2t^2 - 1)] = -1/t + 2t/(2t^2 - 1).$$

Integrate from $t = 1$ (corresponding to $x = 0$) to $t \rightarrow \infty$:

$$\int_1^\infty (-1/t + 2t/(2t^2 - 1)) dt = \lim_{T \rightarrow \infty} [-\ln t + (1/2) \ln(2t^2 - 1)]_1^T = \lim_{T \rightarrow \infty} (1/2) \ln((2T^2 - 1)/T^2) - (1/2) \ln(1) = (1/2) \ln 2. \blacksquare$$

The integral of the dome over its full domain is exactly $\ln(2)/2$. In the language of binary distinction, the dome's total "mass" is half of one bit – the natural measure of the doubling interval on a logarithmic scale.

9.7 The dome's algebra encodes both ingredients of K_{AUD}

Theorems 9.4 and 9.5 together establish a structural fact about the dome function that did not depend on any framework input: the dome's algebra distinguishes both ingredients of $K_{\text{AUD}} = \sqrt{2} \cdot \ln(2)$. Specifically:

(i) The value $\sqrt{2}$ is selected by the dome's self-reference condition $g'/g = g$, with the defining equation $u^2 = 2$ emerging from the dome's logarithmic derivative (Theorem 9.4).

(ii) The value $\ln 2$ is twice the dome's total integral, $\int g dx = \ln(2)/2$, with the integral derived by partial fractions in the substitution $t = \cosh(x)$ (Theorem 9.5).

The dome's own algebra therefore contains both ingredients of K_{AUD} as distinguished algebraic landmarks. What the dome's algebra does not contain is the specific multiplicative combination $\sqrt{2} \cdot \ln(2)$. The product K_{AUD} enters the framework through the construction $K = (\text{geometric primitive}) \times (\text{informational primitive})$ in Paper 3 V3, not through the dome's internal structure. The dome contributes the ingredients; the framework's construction combines them. Whether the multiplicative form (rather than, e.g., an additive or compositional combination of the two primitives) is forced by structural considerations or is one defensible choice among several is the subject of Paper 3 V3's derivation, not of this paper; the present work treats the multiplicative form as given and tests its empirical consequence ($\eta \leq 2G$) at $d = 3$.

This sharpens the structural reading of the empirical bound $\eta \leq 2G$ reported in Part I. The choice of K_{AUD} as the conjectured lower bound on $(2 - \eta)$ is not arbitrary: the value's two ingredients are distinguished landmarks of the dome function that appears in 3D $O(N)$ resummation. Why the dome function appears in $O(N)$ field theory in the first place – the question of whether the Borel resummation has the same saddle-point structure as Hodgson-Kerckhoff tube-packing optimization – is open and discussed in §10.5 below.

9.8 The dilation-operator structure of η_{kernel}

We record a related algebraic fact that connects the dome to the standard kernel sech^2 . Define

the dilation operator on functions of $x \geq 0$ by

$$Df = (1 - x \cdot \partial_x) f(x) = f(x) - x \cdot f'(x).$$

Proposition 9.6. The dome function appears as a dilation image of sech^2 in Mellin space. Specifically, for the Mellin transform $Mf = \int_0^\infty x^{s-1} \cdot f(x) dx$,

$$Mg = (1/2) \cdot M\text{sech}^2 - J(s),$$

where $J(s) = \int_0^\infty x^{s-1} \cdot \text{sech}^2(x) / (e^{4x} + 1) dx$ is a Fermi-Dirac-weighted residual.

Proof. From Theorem 9.1, $g(x) = (1/2) \cdot \text{sech}^2(x) \cdot \tanh(2x)$. Writing $\tanh(2x) = 1 - 2/(e^{4x} + 1)$ and applying the Mellin transform to both terms gives the stated identity directly. ■

At $s = 1$ the decomposition specializes to $\int_0^\infty g dx = (1/2) - J(1)$ where $J(1) = (1/2) - \ln(2)/2$, recovering Theorem 9.5. The residual $J(s)$ admits a closed form (derived in §9.11 as Corollary 9.10 to the complete Mellin spectrum theorem); the $s = 2$ case $J(2) = \ln(2)/2 - \pi^2/32 = 0.038148453\dots$ sits suggestively close to the conformal-bootstrap value of $\eta_{XY} = 0.038176(44)$ (Chester et al. 2020) – within 1σ of the measurement – but is not equal: the algebraic value differs from the central η_{XY} by 2.75×10^{-5} , well-resolved at the bootstrap's actual precision. The proximity is observational; the algebraic value (Corollary 9.10) is not the η_{XY} value.

9.9 The flatness of the corridor in dome coordinates

The three algebraically distinguished landmarks ($\sqrt{2}$, ϕ , 2) of §9.4 are clustered near the dome's peak. The dome rises from 97.5% of its peak value at $u = \sqrt{2}$ to its maximum at $u = \phi$, then falls to 96.1% at $u = 2$ – a full variation of 3.9% below peak across the interval $[\sqrt{2}, 2]$, with the two endpoint values differing from each other by 1.4% of the peak.

u	$g(u)$	$g / g(\phi)$
$\sqrt{2} \approx 1.4142$	$1 - 1/\sqrt{2} \approx 0.2929$	0.975
$\phi \approx 1.6180$	$\phi^{(-5/2)} \approx 0.3003$	1.000 (peak)
2	$1/(2\sqrt{3}) \approx 0.2887$	0.961

The interval $[\sqrt{2}, 2]$ sits entirely within the dome's 96%-of-peak region. The dome's algebraic landmarks therefore coincide with the flat top of the function – the range where $g(u)$ is maximally insensitive to its argument. The "corridor of criticality" in the $(2 - \eta)$ coordinate, which has width $2G \approx 0.0395$ by Fisher's relation and the framework conjecture, corresponds geometrically to this flat-top region in the dome's u -coordinates.

9.10 Two algebraic environments at the HK landmarks

The two Hodgson-Kerckhoff landmarks – $u = 5/4$ (the K_{AUD} -defining position; §12) and $u = 2$ (the critical tube radius R_0 ; §10.3) – sit in distinct exact algebraic environments. At each landmark, all six hyperbolic functions ($\sinh$, $\cosh$, $\tanh$ at x and at $2x$) take exact closed-form values, but the irrational signatures of the two environments differ.

Landmark (I) – at $x = (\ln 2)/2$ (the K_{AUD} -defining position; $u = 5/4$):

Quantity	Exact value
$\sinh(x)$	$1/(2\sqrt{2})$
$\cosh(x)$	$3/(2\sqrt{2})$
$\tanh(x)$	$1/3$
$\sinh(2x)$	$3/4$
$\cosh(2x)$	$5/4$
$\tanh(2x)$	$3/5$

All six values are rational or $\sqrt{2}$ -rational. The dome value here is $g(5/4) = 4/15$ exactly (computable from $g(u) = \sqrt{(u-1)}/[u \cdot \sqrt{(u+1)}]$ at $u = 5/4$, with $\sqrt{(1/4)}/[(5/4) \cdot \sqrt{(9/4)}] = (1/2)/(15/8) = 4/15$). The environment is built on the $\sqrt{2}$ primitive.

Landmark (II) – at $x = R_0 = \text{arccosh}(2)/2$ (the critical tube radius; $u = 2$):

Quantity	Exact value
$\sinh(R_0)$	$1/\sqrt{2}$
$\cosh(R_0)$	$\sqrt{3}/2$
$\tanh(R_0)$	$1/\sqrt{3}$
$\sinh(2R_0)$	$\sqrt{3}$
$\cosh(2R_0)$	2
$\tanh(2R_0)$	$\sqrt{3}/2$

All six values are rational-in- $\sqrt{3}$. The dome value here is $g(2) = 1/(2\sqrt{3})$ exactly. The environment is built on the $\sqrt{3}$ primitive.

Structural reading. The dome's two HK landmarks therefore carry two different irrational geographies, one organized around $\sqrt{2}$ (the framework's named geometric primitive, Paper 3 V3) and one organized around $\sqrt{3}$ (which has no current framework-architectural role at this tier, though $\sqrt{3} = \sinh(2R_0)$ appears as a derived quantity at the binary-doubling point). The two environments meet at the dome's u -axis but are algebraically closed in different irrationals.

This is structurally significant for two reasons. First, it sharpens the §12 "two HK landmarks" discussion: the framework's selection of Landmark (I) over Landmark (II) (the alignment principle of §12.7) is not just a selection between two HK-natural positions, but between two distinct closed algebraic worlds. Second, the appearance of $\sqrt{3}$ as a derived quantity at Landmark (II) is the dome's first generation of an irrational primitive other than $\sqrt{2}$ from its own algebraic structure. Whether $\sqrt{3}$ has additional framework-architectural significance beyond its role at Landmark (II) is an open question; for now we record the observation that the dome has two distinct algebraic environments at its two HK landmarks.

This is an [Observation] tier finding, established by direct computation at $\text{dps} = 50$ verified across multiple architectures; no derivation in any independent setting is currently claimed.

9.11 The Mellin spectrum of the dome – closed form

The dome function admits a complete closed-form expression for all its Mellin moments. This generalizes Theorem 9.5 (the $s = 1$ case $M_g = \ln(2)/2$) and yields a closed form for the residual $J(s)$ of Proposition 9.6 (sharpening §9.8's discussion).

Theorem 9.7 (Mellin spectrum of the dome). For the dome function $g(x) = \tanh(x)/\cosh(2x)$, the Mellin moments

$$M_g = \int_0^\infty x^{s-1} \cdot \tanh(x)/\cosh(2x) \, dx$$

satisfy, for $s \geq 1$:

$$M_g = (s-1)! \cdot \eta(s) \cdot (2^s - 1) / 2^{2s-1}$$

where $\eta(s) = (1 - 2^{1-s}) \cdot \zeta(s)$ is the Dirichlet eta function, with $\eta(1) = \ln 2$ by convention.

Proof. Substitute $y = e^{-(2x)}$, so $x = -(\ln y)/2$, $dx = -dy/(2y)$, and $x^{s-1} = (-1)^{s-1} (\ln y)^{s-1} / 2^{s-1}$. Under this substitution the dome's factorization $g(x) = (1/2) \cdot \text{sech}^2(x) \cdot \tanh(2x)$ (Theorem 9.1) transforms into $g(x) = 2y(1-y)/[(1+y)(1+y^2)]$. Combining,

$$M_g = (-1)^{s-1} \cdot 2^{1-s} \cdot \int_0^1 (\ln y)^{s-1} \cdot (1-y) / [(1+y)(1+y^2)] \, dy.$$

Partial-fraction the rational factor: $(1-y)/[(1+y)(1+y^2)] = 1/(1+y) - y/(1+y^2)$. Using the standard integral $\int_0^1 (\ln y)^k \cdot 1/(1+y) \, dy = (-1)^k \cdot k! \cdot \eta(k+1)$, and the related result (via $u = y^2$) $\int_0^1 (\ln y)^k \cdot y/(1+y^2) \, dy = (-1)^k \cdot k! \cdot \eta(k+1) / 2^{k+1}$,

$$\int_0^1 (\ln y)^{s-1} \cdot (1-y)/[(1+y)(1+y^2)] \, dy = (-1)^{s-1} \cdot (s-1)! \cdot \eta(s) \cdot (1 - 2^{-(s-1)}) = (-1)^{s-1} \cdot (s-1)! \cdot \eta(s) \cdot (2^s - 1) / 2^s.$$

Substituting back gives $M_g = (s-1)! \cdot \eta(s) \cdot (2^s - 1) / 2^{2s-1}$. ■

The case $s = 1$ recovers Theorem 9.5 directly: $0! \cdot \eta(1) \cdot (2 - 1) / 2 = \ln(2)/2$.

Numerical verification. The closed form has been verified at decimal precision $\text{dps} = 1000$ against the rational/ n closed forms for $s = 1, 2, 4, 6, 8$ (Corollary 9.8), with residuals at the working-precision floor (10^{-1000} or identically zero); and against direct numerical integration $\int_0^\infty x^{s-1} g(x) \, dx$ for $s = 1.8$ at $\text{dps} = 500$ (residuals exact or at 10^{-501}) and $\text{dps} = 300$ (residuals at 10^{-300}) – both rounds at the working-precision floor (companion Verification 9.7, re-verified 2026-05-28). The same identity was independently confirmed at $\text{dps} = 1050$ by Grok-Heavy (2026-05-22, zero discrepancies across the full identity set), at machine precision by chat-side Sonnet/Opus (2026-05-25), and at $\text{dps} = 1050$ by Super Grok (2026-06-01, memory-blind to prior framework context, direct $M_g(s)$ computation against the closed form for $s = 1.8$ with residuals identically zero or at the 10^{-1051} working-precision floor).

Corollary 9.8 (even-moment closed forms; $n = 1, 2, 3$ in simplified rational form). Using $\zeta(2n) = |B_{2n}| \cdot (2\pi)^{2n} / (2 \cdot (2n)!)$, Theorem 9.7 specializes to:

n	M_g closed form	Numerical
1	$\pi^2 / 32$	0.30843
2	$7 \pi^4 / 1024$	0.66588
3	$31 \pi^6 / 8192$	3.63807

The numerator coefficients at $n = 1, 2, 3$ are the Mersenne-like values 1, 7, 31, i.e., $(2^{2n-1} - 1)$. This simplified-Mersenne-numerator pattern does NOT extend beyond $n = 3$. At $n = 4$, the closed form gives $M_g = 2159 \pi^8 / 2^{19}$ where $2159 = 17 \cdot 127$, not a single Mersenne factor; the prime 17 enters from the factor $(2^{2n} - 1)$ – at $n = 4$, $2^8 - 1 = 255 = 3 \cdot 5 \cdot 17$ – surviving after cancellation against the $(2n)!$ and $\zeta(2n)$ denominators (which contain only the primes 2, 3, 5, 7 at this order). In the integer-ratio factorization $M(2n)/M(2)^n = A002105[n] \cdot (2^{2n-1} - 1)$ of Corollary 9.9, the prime 17 lands on the A002105 (reduced-tangent-number) side: at $n = 4$, $4318 = 34 \cdot 127 = (2 \cdot 17) \cdot 127$, with the Mersenne-like factor 127 prime and the 17 sitting in the tangent factor $A002105[4] = 34 = 2 \cdot 17$. The general closed form of Theorem 9.7 (with explicit Bernoulli/eta dependence) holds for all n ; the simplified rational form with a single Mersenne numerator is a small- n property that breaks at $n \geq 4$. We list the $n = 1, 2, 3$ cases here for the structurally clean part of the pattern and direct readers to Theorem 9.7 for general n .

Corollary 9.9 (integer ratio sequence $M(2n)/M(2)^n$). The dimensionless ratio M_g/M_g^n is an integer for all $n \geq 1$:

$$Mg / Mg^n = (2^{(2n-1)} - 1) \cdot (2^{(2n)} - 1) \cdot |B_{(2n)}| \cdot 2^n / n.$$

n	M(2n)/M(2)^n
1	1
2	7
3	124
4	4318
5	253456
6	22631632
7	2862787264
8	487346998768

The Bernoulli prefactor $|B_{(2n)}| \cdot 2^n / n$ equals the reduced tangent number a_n (OEIS A002105 = 1, 1, 4, 34, 496, 11056, 349504, 14873104, ...), coming from the Taylor series of $\tan(x)$. [Erratum, 2026-09-11: the sentence above drops a factor. The prefactor that equals A002105(n) is $(2^{(2n)} - 1) \cdot |B_{(2n)}| \cdot 2^n / n$ – at $n = 1$ the bare prefactor is $1/3$, not 1; at $n = 4$ it is $2/15$, not 34. Theorem 9.7 and Corollary 9.9 stand exactly as printed and agree for all n (exact arithmetic, $n = 1..8$); Paper 12 states the relation correctly. Found 2026-08-26 (D.B.), verified the same hour (Opus); no computed value and no theorem is affected.] Therefore the dome's even-moment ratio sequence factorizes as a Theorem-tier algebraic identity:

$$Mg / Mg^n = A002105(n) \cdot (2^{(2n-1)} - 1) \text{ for all } n \geq 1.$$

Verified at $n = 1..8$ by independent subagent search of OEIS (the exact product sequence is not in OEIS as a direct entry, but the factorization into A002105 and the Mersenne-like sequence is exact).

Observation (§9.11.4): structural reading of the factorization. The dome's algebraic factorization $g(x) = (1/2) \cdot \text{sech}^2(x) \cdot \tanh(2x)$ (Theorem 9.1) decomposes the function into a Lorentzian spectral envelope (sech^2) and a binary-doubling carrier ($\tanh(2x)$). The Theorem 9.7 closed form, combined with the A002105 $\times$ Mersenne factorization of Corollary 9.9, suggests an interpretation: the reduced tangent numbers A002105 are the tangent-class component of the moment arithmetic, structurally aligned with the $\tanh(2x)$ factor; the Mersenne factors $(2^{(2n-1)} - 1)$ are the binary-doubling component, structurally aligned with the $\cosh(2x)$ denominator. This is a parallel observation about two distinct factorizations both existing – one at the function level, one at the moment-arithmetic level – both decomposing along similar structural lines.

This reading is Observation-tier, not Theorem-tier. The algebraic identity of Corollary 9.9 is a theorem; the interpretation that each factor of the integer-ratio sequence corresponds to a specific component of the dome's algebraic factorization is a structural observation that has not been derived from the dome's functional decomposition. Promotion to Theorem would require an independent derivation of the two integer families from the two factor components of $g(x)$, which has not been built. We label this Observation explicitly to maintain the framework's Λ -canary discipline against conflating algebraic identities with their structural interpretations. This promotion path is recorded as a candidate Tier 6 derivation target in §11.5(f).

Corollary 9.10 (closed form for the residual $J(s)$ of Proposition 9.6). Combining Theorem 9.7 with the known closed form $M\text{sech}^2 = 2^{(2-s)} \cdot \Gamma(s) \cdot \eta(s-1)$ yields a closed form for the Fermi-Dirac-weighted residual of Proposition 9.6:

$$J(s) = (s-1)! \cdot [2^s \cdot \eta(s-1) - (2^s - 1) \cdot \eta(s)] / 2^{(2s-1)}$$

In particular, the $s = 2$ case (recorded in §9.8 of this paper as the numerical near-miss $J(2) \approx 0.038148$ vs $\eta_{XY} \approx 0.038176$) acquires its exact algebraic form:

$$J(2) = \ln(2)/2 - \pi^2 / 32 \text{ (exact)}$$

This sharpens the discussion of §9.8: $J(2)$ is the specific algebraic value $\ln(2)/2 - \pi^2/32$, and η_{XY} at conformal-bootstrap precision is approximately but not equal to this value. The "not equal" framing of §9.8 is preserved; what changes is that the value now has a name – it is no longer "an integral with no known closed form whose numerical value is 0.038148..."; it is the rational/ π combination $\ln(2)/2 - \pi^2/32$ exactly. Verified at $\text{dps} = 1000$ (residual identically zero between $J(2)$ and $\ln(2)/2 - \pi^2/32$ at the working-precision floor); companion Verification 9.7.

9.12 What §9 establishes

Returning to the master landmark table at the start of §9, we can now state what the dome's algebraic architecture establishes:

(a) Its peak is at the golden ratio ϕ exactly (Corollary 9.3). (b) It distinguishes $\sqrt{2}$ as the unique self-referential fixed point of its logarithmic-derivative-equals-value condition (Theorem 9.4). (c) Its total integral equals $\ln(2)/2$ exactly (Theorem 9.5) – equivalently, twice its total mass is $\ln 2$, the framework's informational primitive. (d) Its two HK landmarks ($u = 5/4$ and $u = 2$) sit in distinct exact algebraic environments – Landmark (I) $\sqrt{2}$ -rational, Landmark (II) $\sqrt{3}$ -rational (§9.10). (e) Its complete Mellin spectrum admits a closed form (Theorem 9.7): $Mg = (s-1)! \cdot \eta(s) \cdot (2^s - 1) / 2^{(2s-1)}$. Even moments are rational multiples of $\pi^{(2n)}$; the integer ratio sequence $M(2n)/M(2)^n$ factorizes as the reduced tangent numbers A002105 times the Mersenne-like factors (Corollary 9.9). Proposition 9.6's Fermi-Dirac residual $J(s)$ acquires a closed form by combination (Corollary 9.10), with $J(2) = \ln(2)/2 - \pi^2/32$

exactly. (f) Together with the corridor identity $2 - 2K_{AUD} = 2G$ (§8.1) and the corridor margin $G/\ln 2 = G \cdot \log_2(e)$ (Theorem 13.1), the dome's geometry implies the algebraic constraints on η that Part I records empirically.

These properties are independent of any framework input. They are facts about the function. Their relevance to the empirical observation of Part I is mediated by the fact, established in §10 below, that the same function appears as the resummation factor $R(N)$ for the anomalous dimension at the 3D Wilson-Fisher fixed point, with the framework's primary constants $\{\phi, \sqrt{2}, 2\}$ populating the algebraic landmarks in both domains.

The framework's two primary constants – $\sqrt{2}$ (geometric) and $\ln 2$ (informational) – are generated by the dome's two internal structures (self-reference and integral). Their product $K_{AUD} = \sqrt{2} \cdot \ln 2$ enters the framework architecture as a multiplicative combination of two landmarks the dome distinguishes by its own algebra – but, as §9.7 records, the dome's algebra does not contain the multiplicative combination $\sqrt{2} \cdot \ln(2)$ itself: the dome contributes the ingredients separately, and the multiplicative combination enters through the framework's construction $K = (\text{geometric primitive}) \times (\text{informational primitive})$ in Paper 3 V3. Both $\sqrt{2}$ and $\ln 2$ are ubiquitous mathematical constants individually, so their co-appearance in the dome's two unrelated internal conditions is suggestive of structural relevance, not strong evidence by itself. The dome's complete Mellin spectrum (Theorem 9.7) further establishes that the dome carries the framework's primitives not only at the function level but throughout its moment arithmetic. This is the structural reading the master table at the start of §9 carries: the dome's algebra contains the framework's primitives at the ingredient level – with the multiplicative combination supplied by the framework's construction – and the framework's other findings sit on the dome's landmarks.

10. The dome in two domains: Hodgson-Kerckhoff tube-packing and 3D $O(N)$ resummation

10.1 The Hodgson-Kerckhoff tube-packing function

Hodgson and Kerckhoff (2005), in their work bounding hyperbolic Dehn surgery on cusped 3-manifolds, derive the tube-packing function

$$h(r) = 2\sqrt{6} \cdot \ln(2) \cdot \tanh(r) / \cosh(2r), \quad r \geq 0,$$

where r is the tube radius in a hyperbolic 3-manifold. The function bounds the volume of an embedded geodesic tube and enters the proof of universal bounds on the change of geometric data under Dehn filling. Comparing to §9, $h(r)$ is the dome function $g(r)$ of Theorem 9.1 multiplied by the constant $2\sqrt{6} \cdot \ln(2)$. The factor $2\sqrt{6} \cdot \ln(2)$ is the maximum value of $h \times$ something else in HK's normalization; the dimensionless dome shape is unchanged.

The function is published in [Hodgson-Kerckhoff 2005] and used in subsequent hyperbolic-geometry work [Futer-Purcell-Schleimer 2022]. Within hyperbolic geometry it is a known object. The algebraic structure of §9 – the golden-polynomial logarithmic derivative, the self-reference at $\sqrt{2}$, the integral identity – was not stated in the original HK paper or any citing work we have located; it remained one calculation away from the published literature for the two decades between HK's publication and the present.

10.2 The peak at ϕ – same algebraic fact, both domains

By Corollary 9.3, the dome function peaks at $u = \cosh(2r) = \phi$. Substituting into HK's function gives the explicit peak value:

$$h_{\text{peak}} = 2\sqrt{6} \cdot \ln(2) \cdot \phi^{(-5/2)}.$$

The HK dome peaks at the golden ratio exactly, as an algebraic consequence of the golden polynomial in the logarithmic derivative (Theorem 9.2). This is the same algebraic fact as in §9 – the HK function and the dome function differ only by the multiplicative constant $2\sqrt{6} \cdot \ln(2)$ – but it is worth noting that the algebraic location of the peak was not recognized in the hyperbolic-geometry literature before its rediscovery in the framework's analysis. The peak condition $u^2 - u - 1 = 0$, in which ϕ is the unique positive root, follows immediately from setting the logarithmic-derivative numerator of Theorem 9.2 equal to zero.

10.3 The critical tube radius R_0 and the algebraic identity $h(R_0) = K_{AUD}$

Define the critical tube radius by

$$R_0 = \operatorname{arccosh}(2) / 2,$$

equivalently $R_0 = \operatorname{arctanh}(1/\sqrt{3}) = \operatorname{arcsinh}(1/\sqrt{2})$. All three identities follow from the relation $\cosh(2R_0) = 2$ by elementary hyperbolic identities. Numerically $R_0 \approx 0.6585$.

At $r = R_0$ the dome takes a structurally distinguished value:

- $\cosh(2R_0) = 2$ (the binary doubling threshold; the value $u = 2$ is the unit-polynomial landmark $P(u) = 1$ of §9.4).
- $\tanh(R_0) = 1/\sqrt{3}$ (from $\operatorname{arctanh}$ form).
- $\sinh(R_0) = 1/\sqrt{2}$ (from $\operatorname{arcsinh}$ form).
- $\cosh(R_0) = \sqrt{3/2}$.

Substituting into HK's function:

$$h(R_0) = 2\sqrt{6} \cdot \ln(2) \cdot (1/\sqrt{3}) / 2 = \sqrt{6} \cdot \ln(2) / \sqrt{3} = \sqrt{2} \cdot \ln(2) = K_{\text{AUD}}.$$

Theorem 10.1. $h(R_0) = K_{\text{AUD}}$ exactly.

The identity is equivalent to the elementary hyperbolic identity $\text{arcsinh}(1/(2\sqrt{2})) = \ln(2)/2$, which can be verified by computing $\sinh(\ln(2)/2) = (e^{(\ln(2)/2)} - e^{-(\ln(2)/2)})/2 = (\sqrt{2} - 1/\sqrt{2})/2 = (1/(2\sqrt{2})) \cdot (2 - 1) = 1/(2\sqrt{2})$. Verification in HK_Closed_Form.txt and HK_Verification_Log.txt (D.B., April 2026) at multiple precisions across multiple architectures.

The structural reading: the HK function, at its critical tube radius (the binary-doubling point of the dome's u-coordinate), takes the value K_{AUD} . The constant K_{AUD} therefore appears in the hyperbolic-geometry literature as the value of a published tube-packing function at a structurally distinguished radius – without any framework input.

10.4 The descent ratio from peak to critical radius

The dome's descent from its peak (at $u = \phi$) to the critical tube radius (at $u = 2$) is exact:

$$h(R_0) / h_{\text{peak}} = [2\sqrt{6} \cdot \ln(2) / 2 \cdot (1/\sqrt{3})] / [2\sqrt{6} \cdot \ln(2) \cdot \phi^{(-5/2)}] = \phi^{(5/2)} / (2\sqrt{3}) \approx 0.96134.$$

The dome descends 3.866% from its peak to the binary-doubling point – equivalently, it retains 96.134% of its peak value. This is the flat top property identified abstractly in §9.9; here we see it concretely between two algebraically distinguished landmarks. The interval $[\phi, 2]$ in u-coordinates is the flat top of the dome; the HK function loses less than 4% of its peak value over this entire interval.

10.5 The same function appears in 3D $O(N)$ resummation

We now turn to the other domain. At the 3D Wilson-Fisher fixed point, the anomalous dimension $\eta(N)$ for the $O(N)$ universality class is, in leading-order ϵ -expansion,

$$\eta_2(N) \cdot \epsilon^2 = (N + 2) / [2(N + 8)^2] \cdot \epsilon^2,$$

with $\epsilon = 4 - d = 1$ in $d = 3$. The full all-order η is reached from η_2 by Borel resummation of the perturbative series, whose factorial growth is controlled by the leading Borel singularity

$$a(N) = 3 / (N + 8)$$

(the Lipatov-instanton position; standard reference: Brézin–Le Guillou–Zinn-Justin 1977 and McKane–Wallace; persistence at all orders verified to 7-loop in Dunne–Meynig 2021/2023, arXiv:2111.15554). The dimensionless resummation factor $R(N) = \eta(N) / [\eta_2(N) \cdot \epsilon^2]$ measures how much the full all-order η exceeds its leading-order ϵ^2 value.

We propose the following zero-parameter model:

$$\text{Model 10.2. } R(N) = 2\phi^{(5/2)} \cdot \tanh(c \cdot a(N)) / \cosh(2c \cdot a(N)), \text{ with } c = 2 \ln(2), a(N) = 3 / (N + 8).$$

The model has the same functional form as the HK tube-packing function of §10.1 (up to overall amplitude), with the Borel singularity $a(N)$ playing the role of the tube radius r and the framework constant $c = 2 \ln 2$ setting the hyperbolic scale.

Free parameters: zero. Both the amplitude $A = 2\phi^{(5/2)} \approx 6.6604$ and the scale $c = 2 \ln(2) \approx 1.3863$ are framework constants. The Borel singularity $a(N)$ is fixed by the physics. The amplitude is fixed by the requirement $R_{\text{peak}} = 2$ (the topological doubling limit; see §10.6 below). The scale c is the alignment-principle choice between two HK landmarks at $d_c = 4$, discussed in §12.

Comparison with data (3D $O(N)$ η values from conformal bootstrap and Monte Carlo, as cited in Part I §5):

Class	N	R_data	R_model	Relative error
SAW	0	1.987	1.999	+0.63%
Ising	1	1.960	1.972	+0.61%
XY	2	1.909	1.918	+0.48%
Heisenberg	3	1.844	1.850	+0.33%
$O(4)$	4	1.752	1.776	+1.38%

The root-mean-square relative error is 0.77% across $N = 0-4$ with zero free parameters. A best-2-parameter fit gives $A_{\text{fit}} = 6.631$ (0.44% from the framework value) and $c_{\text{fit}} = 1.375$ (0.80% from the framework value); the framework values sit within these tight tolerances of the data-best-fit values.

10.6 $R_{\text{peak}} = 2$ as an algebraic consequence

The maximum of the dome $g(u)$ at $u = \phi$ is $g(\phi) = \phi^{(-5/2)}$. Therefore the model's maximum value is

$$R_{\text{peak}} = 2\phi^{(5/2)} \cdot \phi^{(-5/2)} = 2 \text{ (exact).}$$

The amplitude $A = 2\phi^{(5/2)}$ was chosen specifically to fix $R_{\text{peak}} = 2$; this is the topological doubling limit, the boundary above which the resummation factor would exceed unity-times-two

and the perturbative series would be "less than half right" in a precise sense. The peak of the model is reached at $N \approx -0.16$, slightly below the physical range $N \geq 0$; for all physical $O(N)$ models the model predicts $R(N) < 2$, with the closest approach at $N = 0$ (SAW) where $R_{\text{model}} = 1.999$. The " $R < 2$ " property is therefore not an empirical accident but a built-in consequence of the amplitude matching.

10.7 Caveats on the zero-parameter fit

The 0.77% RMS fit at $N = 0-4$ is striking but requires honest qualification:

- (a) Five smooth monotonic data points. $R(N)$ is monotonically decreasing across $N = 0-4$. Any smooth two-parameter function fitting five monotonic points will typically achieve sub-1% accuracy. The fact that the model's zero-parameter version is also sub-1% accurate is more constraining than the best-fit 2-parameter version, but five points is a small sample.
- (b) The model degrades beyond $N = 4$. Extension to $N = 5, 6, 8, 10, 16, 32$ (via $1/N$ expansion, functional RG, and Monte Carlo at increasing N) shows the model's relative error growing: $\sim 3\%$ at $N = 5-6$, $\sim 9\%$ at $N = 16$, $\sim 30\%$ at $N = 32$. The model cannot reproduce the large- N asymptote $R(N) \rightarrow 16/(3\pi^2) \approx 0.540$ (Vasiliev-Pismak-Khonkonen 1981) because $R_{\text{model}}(N) \rightarrow 0$ as $a(N) \rightarrow 0$. The model is a local description of the dome's peak structure in the physical range, not a universal RG-theoretic resummation formula.
- (c) Leading-order Borel singularity only. The model uses only the leading Lipatov-instanton position $a(N) = 3/(N + 8)$. Subleading saddle points (the four-instanton tower, renormalon contributions in certain RG schemes) are not included. The systematic positive bias of the model errors (all five entries overshoot R_{data}) suggests a missing monotone correction, possibly from subleading Borel structure.
- (d) The empirical bound is more robust than the model. At every N tested up to $N = 32$, the ratio $\eta/(2G)$ remains below 0.97 – the bound $\eta \leq 2G$ itself is preserved across a range where the dome-model fit fails. The bound is therefore a more fundamental statement than the dome shape that argues for it locally. This points toward a derivation of $\eta \leq 2G$ that does not necessarily go through the HK dome form. The dome form motivates the choice of K_{AUD} as the ceiling-setting constant in the physical range where the model fits; the bound's empirical universality survives outside that range.

10.8 The triple convergence at R_0

The critical tube radius $R_0 = \text{arccosh}(2)/2$ of §10.3 emerges as a distinguished hyperbolic position in three independent settings:

Setting	What R_0 is in this setting	Defining identity
Hodgson-Kerckhoff tube packing	Binary-transition tube radius where $h(R_0) = K_{\text{AUD}}$	$\cosh(2R_0) = 2$ (HK 2005)
η -resummation dome	Max-dilation of $\eta_{\text{kernel}} = D[\text{sech}^2]$	$\text{sech}^2(R_0) = 2/3$
Non-rotating BTZ (2+1 dimensions)	Half-geodesic horizon $\rightarrow r = 2r_h$	$r/r_h = \sqrt{3/2}$

All three identifications reduce algebraically to $\cosh(2R_0) = 2$ by elementary hyperbolic identities. The triple convergence is verified at $\text{dps} = 100$ in the framework's verification logs (sommerfeld_verify.py). What it establishes is that R_0 is a structurally distinguished location in three different hyperbolic-geometric constructions; what it does not establish is a shared derivation explaining why these three constructions converge on the same value. The convergence is observed, not derived; the open question is whether there is an underlying mechanism that forces R_0 to appear in all three settings, or whether the appearances reflect a deeper structural identity (a "Topological Bound on Information Diffusion" or analogous unifying principle) that has not yet been formulated.

10.9 The deeper structural question – Borel-hyperbolic map as derivation target

The model of §10.5 fits 3D $O(N)$ data with zero free parameters and matches the HK tube-packing function up to overall amplitude. But the model is a fit, not a derivation. The structural question that the parallel raises is the following.

Why does the 3D $O(N)$ ϕ^4 resummation factor $R(N)$ have the hyperbolic-doubling kernel form in the first place?

A derivation would require showing that the resummation integral over the Borel plane in $O(N)$ ϕ^4 theory has the same saddle-point structure as the Hodgson-Kerckhoff tube-packing optimization in hyperbolic 3-manifold geometry. Specifically, the conjectural derivation target is:

Show that the Borel singularity $a(N) = 3/(N + 8)$ plays the role of the tube radius r in an optimization problem whose Euler-Lagrange equations coincide with those of HK tube packing.

This is the framework's central open Tier 6 question for the η -corridor. If proved, the dome's appearance in $O(N)$ resummation becomes forced (not coincidental), the framework constants populating its landmarks become forced rather than empirically matched, and the empirical bound $\eta \leq 2G$ acquires a derivation route through:

1. The dome's own algebra produces $\sqrt{2}$ and $\ln(2)/2$ as structurally distinguished landmarks (Theorems 9.4 and 9.5).
2. The framework's construction $K = (\text{geometric primitive}) \cdot \ln 2$ combines them into K_{AUD} (Paper 3 V3).
3. Fisher's relation $\gamma = \sqrt{2} - \eta$ doubles K_{AUD} into the corridor ceiling $2K_{\text{AUD}}$ (Part I

§8.1).

4. The Borel-hyperbolic map shows the dome's appearance in $O(N)$ resummation is forced by saddle-point structure (open target).

Steps 1–3 are in hand. Step 4 is the bridge.

10.10 What §10 establishes and does not establish

Established (Theorem-tier or Observation-tier with high-precision verification):

- The Hodgson-Kerckhoff tube-packing function (published 2005) takes the form of the dome function up to an overall amplitude $\sqrt{6} \cdot \ln(2)$ (§10.1).
- The HK function peaks at the golden ratio ϕ exactly (§10.2, Corollary 9.3).
- The HK function at the critical tube radius R_0 takes the value $K_{AUD} = \sqrt{2} \cdot \ln(2)$ exactly (Theorem 10.1).
- The dome function appears as a zero-parameter model for the 3D $O(N)$ resummation factor $R(N)$ at 0.77% RMS in the range $N = 0-4$, with the framework constants $\{\phi, \sqrt{2}, 2\}$ populating its algebraic landmarks (Model 10.2 and the comparison table of §10.5).
- The critical tube radius R_0 appears as a structurally distinguished hyperbolic position in three independent constructions (§10.8).

Not established (open Tier 6 questions):

- Why the dome function appears in 3D $O(N)$ Borel resummation at all (the Borel-hyperbolic-map conjecture, §10.9).
- Whether the triple convergence at R_0 reflects a shared underlying mechanism or three independent constructions converging on the same value (§10.8).
- Whether the framework's specific scale $c = 2 \ln 2$ (rather than the alternative HK-landmark choice $c = 2 \cdot \operatorname{arcsinh}(1/\sqrt{2}) \approx 1.762$) is structurally forced or empirically selected. The discussion of this choice is the subject of §12.

11. The Central Synchronization Conjecture

11.1 Why this section exists

$K_{AUD} = \sqrt{2} \cdot \ln(2)$ has appeared, with no free parameters and to numerical precision verified at $dps = 1500$ across multiple architectures, in several mathematical settings. Holding the framework's A-canary discipline (multi-mechanism convergence requires distinct mechanisms, not multiple realizations of the same algebraic identity), these settings sort into one hyperbolic-kernel cluster routing through a shared algebraic identity, plus three mechanism-distinct anchors, plus a related ingredient-level finding.

Hyperbolic-kernel cluster (one shared algebraic identity, three operational contexts). The identity

$$\operatorname{arcsinh}(1/(2\sqrt{2})) = (\ln 2)/2, \text{ equivalently } \sinh((\ln 2)/2) = 1/(2\sqrt{2}),$$

is the algebraic kernel through which K_{AUD} enters three otherwise unrelated constructions:

- The Hodgson-Kerckhoff tube-packing function in hyperbolic 3-manifold geometry (HK 2005, Theorem 4.4; §10.3 of this paper), where the identity gives the closed form $1/S = \sqrt{2} \cdot \ln 2 = K_{AUD}$ inside HK's bound machinery.
- The saturation boundary of Algorithm EA at the binary cell $L = \ln 2$ (Saturation Constants paper, §3), where the identity provides the closed-form residual probability $p(\ln 2) = \ln 2 / (2 \cdot \sinh((\ln 2)/2)) = \sqrt{2} \cdot \ln 2 = K_{AUD}$.
- The dome's evaluation at the upper critical dimension $d_c = 4$ (§12.4 of this paper), which lands at the position $x = (\ln 2)/2 = \operatorname{arcsinh}(1/(2\sqrt{2}))$ – explicitly the same algebraic point as the HK landmark, in different vocabulary.

These three are operationally distinct (hyperbolic 3-manifold geometry, Monte Carlo sampling endpoint-balancing, Borel-resummation hyperbolic coordinate at d_c) but algebraically identical: they realize one identity in three different operational settings. Counted under the framework's A-canary discipline, they constitute one convergence on the same algebraic fact, not three independent appearances.

Mechanism-distinct anchors (no shared hyperbolic-kernel identity).

- The H_4 regular polytope (Paper 3 V3, DOI: 10.17605/OSF.IO/QH5S2), where $\sqrt{2}$ appears as the framework's geometric primitive associated with H_4 . Paper 3 V3 §3.4 catalogues four origins for the $\sqrt{2}$ factor (the $\sqrt{2}/2$ factor in the 120-cell circumradius formula $R = (\sqrt{2}/2) \cdot \sqrt{5 + \sqrt{5}} \approx 1.9021$; the $\sqrt{2}$ scaling factor in the L2 norm of an equal-component 2-vector, where $\|(\ln 2, \ln 2)\|_2 = \sqrt{2} \cdot \ln 2$ and the $\sqrt{2}$ is the norm-scaling factor, not the norm itself; hypercube body-diagonal geometry; the algebraic root of $x^2 - 2$), with no commitment to a single privileged origin. The framework constructs K_{AUD} by combining $\sqrt{2}$ with the informational primitive $\ln 2$. No $\operatorname{arcsinh} / \sinh$ identity is involved; the polytope-geometry route is structurally independent of the hyperbolic-kernel cluster.
- The threshold coefficient in an upper bound on the Erdős–Hooley Δ -function average order (de la Bretèche–Tenenbaum 2022, arXiv:2210.13897), where the value $\sqrt{2} \cdot \log 2 \approx 0.980258$ appears as the threshold coefficient a in an upper bound $\Delta^*(x) \leq C \cdot \exp(a \cdot \sqrt{\log_2 x})$ on the (weighted) average order of the Δ -function. Exponential-sum / Dirichlet-series saddle-point analysis; no hyperbolic kernel.
- The empirical ceiling on the anomalous dimension η at the 3D Wilson-Fisher fixed point

(Part I of this paper), where K_{AUD} appears via Fisher's relation as the corridor's lower endpoint $2K_{AUD}$ in the $(2 - \eta)$ coordinate. Empirical observation across ten WF-family classes; counts as independent evidence at the empirical-tier level rather than as a derivation.

Ingredient-level finding (different kind of evidence). Independently of the above, the dome function studied in §9 – defined with no reference to H_4 polytope geometry, to the EA algorithm, to the framework, or to any of the above derivations – produces both ingredients of K_{AUD} as algebraically distinguished landmarks ($\sqrt{2}$ from the self-reference condition $g'/g = g$, with the equation reducing to $u^2 = 2$; $\ln 2$ as twice the function's total integral). The dome's algebra does not contain the specific multiplicative combination $\sqrt{2} \cdot \ln(2)$ itself (§9.7); the product enters through the framework's construction in Paper 3 V3, not through the dome. The dome's contribution is therefore ingredient-level evidence: both factors are distinguished by the dome's internal structure, but the multiplicative combination is supplied by framework construction. Both $\sqrt{2}$ and $\ln 2$ are ubiquitous mathematical constants individually, so their co-appearance in the dome's two unrelated internal conditions is suggestive, not strong evidence by itself.

The reader's natural question is whether the hyperbolic-kernel cluster, the three mechanism-distinct anchors, and the ingredient-level finding represent independent coincidences clustering at the same numerical value, or whether they are the same structural object expressing itself in different mathematical clothing. The Central Synchronization Conjecture states the second reading explicitly, while §11.6 acknowledges the first reading as logically possible.

11.2 The $n = 2$ sub-unity selection theorem

Before stating the conjecture, we record a theorem that underlies its formulation.

Theorem 11.1 ($n = 2$ sub-unity selection). Define $K(n) = \sqrt{n} \cdot \ln(n)$ for positive integers n . Then $K(n) < 1$ if and only if $n = 2$. Specifically: $K(1) = 0$, $K(2) = \sqrt{2} \cdot \ln 2 \approx 0.9803$ (the unique sub-unity value at integer $n > 1$), $K(3) = \sqrt{3} \cdot \ln 3 \approx 1.9027$, and $K(n)$ is monotonically increasing for $n \geq 2$.

Proof. $K(2) = \sqrt{2} \cdot \ln 2$; numerical evaluation gives $K(2) \approx 0.9803 < 1$. $K(3) = \sqrt{3} \cdot \ln 3 \approx 1.9027 > 1$. For $n \geq 3$, $d/dn[\sqrt{n} \cdot \ln n] = (\ln n + 2)/(2\sqrt{n}) > 0$, so K is monotonically increasing on $n \geq 3$. $K(1) = \sqrt{1} \cdot \ln 1 = 0$. So K crosses unity exactly once on positive integers, between $n = 2$ and $n = 3$, with $n = 2$ being the unique positive integer for which $0 < K(n) < 1$. ■

Verified across architectures at $dps = 1500$ (atlas lock `n2_sub_unity_selection_theorem`).

Structural reading. The function $K(n) = \sqrt{n} \cdot \ln n$ combines a geometric primitive ($\sqrt{n}$, the H_n -family circumradius pattern) with an informational primitive ($\ln n$, the Shannon entropy of the unit n -ary distinction). The product crosses unity exactly once on positive integers, with $n = 2$ being uniquely sub-unity. The value $K(2) = K_{AUD}$ is therefore the distinguished value of the geometry-information product at the unique sub-unity integer.

11.3 Statement of the conjecture

Conjecture 11.2 (Central Synchronization Conjecture). $K_{AUD} = \sqrt{2} \cdot \ln(2)$ is the value of the geometry-information intersection at the unique sub-unity integer $n = 2$. The conjecture has two parts:

(i) **Structural claim.** There is a single underlying mathematical object – the geometry/information intersection at $n = 2$ – that K_{AUD} is the value of. The recurrences of K_{AUD} across (a) the hyperbolic-kernel cluster of HK tube-packing + Algorithm EA + dome-at-d_c (sharing the algebraic identity $\operatorname{arcsinh}(1/(2\sqrt{2})) = (\ln 2)/2$ per §11.1), (b) the three mechanism-distinct anchors of the H_4 polytope, the Erdős-Hooley threshold, and the Wilson-Fisher η -ceiling, and (c) the dome's ingredient-level finding (the dome's algebra distinguishes $\sqrt{2}$ and $\ln 2$ separately but not their product) are not independent.

(ii) **Participation claim.** Each domain where K_{AUD} has been derived participates in this intersection through a domain-specific mechanism (H_4 closure, EA saturation, HK packing, the Tenenbaum threshold, the Wilson-Fisher fixed point). These mechanisms are not arbitrary – each one realizes the $n = 2$ sub-unity geometry/information intersection in its domain's native variables.

The conjecture is Tier 5/6 boundary: its supporting items (§11.4 below) are individually at Theorem-tier or Observation-tier with high-precision verification, but the overarching structural claim that they are the same intersection has not been derived. The Tier 6 work that would promote the conjecture to theorem is named in §11.5.

11.4 Partial supporting theorems and observations

The following items are individually at Theorem-tier or Observation-tier with high-precision verification ($dps = 1050$ or higher across multiple architectures, unless otherwise noted). They collectively constitute the partial-evidence base for Conjecture 11.2.

Item 1 (Theorem). The $n = 2$ sub-unity selection theorem (Theorem 11.1 above). $K(n) = \sqrt{n} \cdot \ln n$ is uniquely sub-unity at $n = 2$ among positive integers.

Item 2 (Theorem). The dome's algebra encodes both K_{AUD} ingredients. The dome function $g(x) = \tanh(x)/\cosh(2x)$, in $u = \cosh(2x)$ coordinates, distinguishes both $\sqrt{2}$ (via the self-reference condition $g'/g = g$, with the equation reducing to $u^2 = 2$; Theorem 9.4) and $\ln(2)/2$ (as the

function's total integral $\int g \, dx$; Theorem 9.5). The dome's own algebra contains the raw material for $K_{AUD} = \sqrt{2} \cdot \ln 2$ without any framework input.

Item 3 (Theorem). Triple convergence at R_0 . The critical hyperbolic radius $R_0 = \text{arccosh}(2)/2$ appears as a structurally distinguished position in three independent constructions: the HK tube-packing function (binary-transition radius where $h(R_0) = K_{AUD}$), the η -resummation dome (maximum-dilation point of $\eta_{\text{kernel}} = D[\text{sech}^2]$), and the non-rotating BTZ black hole in 2+1 dimensions (half-geodesic-distance from horizon to doubling radius). All three identifications reduce algebraically to $\cosh(2R_0) = 2$ (§10.8).

Item 4 (Theorem). The corridor identity $2 - 2K_{AUD} = 2G$. Given $K_{AUD} + G = 1$, the corridor width $2G$ is algebraically forced by Fisher's relation $\gamma = v(2 - \eta)$ supplying the upper endpoint of $(2 - \eta)$ and the framework's conjectured lower endpoint $2K_{AUD}$. The empirical bound $\eta \leq 2G$ holds across all measured 3D Wilson-Fisher-family classes (Part I §5, with the tightest unitary case at +2.20).

Item 5 (Theorem). The dilation-operator structure of η_{kernel} . The kernel $\eta_{\text{kernel}} = D[\text{sech}^2]$, where $D = 1 - x \cdot \partial_x$ is the dilation generator, realizes the same algebraic structure in three coordinate views: as a Mellin transform with $(s + 1)$ as the dilation signature, as the dome function in $\cosh(2x)$, and as a packing functional in the hyperbolic-geometry setting. Three independent derivations from three different starting points produce the same kernel.

Item 6 (Observation). The zero-parameter HK form of $R(N)$. The Borel-resummation factor $R(N)$ for 3D $O(N)$ ϕ^4 theory across $N = 0-4$ fits the Hodgson-Kerckhoff tube-packing function at 0.77% RMS with zero free parameters, with the framework constants $\{\phi, \sqrt{2}, 2\}$ populating the function's algebraic landmarks (§10.5). The same dome function appears in HK and in $O(N)$ resummation.

Item 7 (Observation). H_4 at $d = 4$ and $d_c = 4$ coincidence. The H_4 regular polytope (source of $\sqrt{2}$ via circumradius) lives in dimension $d = 4$. The upper critical dimension for $O(N)$ ϕ^4 theory established by Wilson and Fisher (1972) is $d_c = 4$. These are the same dimension from independent sources (geometric vs renormalization-group). The bridge between H_4 at $d = 4$ and the Wilson-Fisher fixed point at $d_c = 4$ has not been built; the coincidence is recorded as an observation.

Item 8 (Observation, with one Theorem-tier component). The $N = d_c = 4$ binary landmark alignment. At the upper critical dimension $N = d_c = 4$, three algebraically independent inputs align on the dome's binary landmark $u = 5/4$: (i) the Lipatov-instanton position $a(d_c) = 1/d_c$ at all orders (Theorem; verified to 7 loops via Dunne-Meynig 2021/2023, arXiv:2111.15554); (ii) the framework's scale $c = (d_c/2) \cdot \ln 2$ placing the dome at the K_{AUD} -defining hyperbolic position $\text{arcsinh}(1/(2\sqrt{2})) = \ln(2)/2$ (Theorem-conditional on the alignment principle; see §12); (iii) the elementary hyperbolic identity $\cosh(\ln 2) = 5/4$. The alignment is exact; the structural reading depends on the alignment principle's status, discussed in §12.

Items 1–8 are individually at Theorem or Observation tier with verification. They are not assertions; they are mathematical facts about specific functions, polynomials, integrals, and empirical fits.

11.5 What is conjectural (Tier 6 work remaining)

The conjecture's structural claim – that items 1–8 are not eight independent coincidences but eight aspects of a single intersection – is the part that remains to be proved. The specific Tier 6 derivation targets that would promote Conjecture 11.2 from conjecture to theorem are:

(a) The Borel-hyperbolic map (§10.9). Show that the Borel singularity $a(N) = 3/(N + 8)$ of $O(N)$ ϕ^4 theory plays the role of the tube radius in an optimization problem with the same saddle-point structure as HK tube-packing. If proved, Item 6 becomes a derivation rather than an empirical fit.

(b) The EA saturation participation map. Show that Algorithm EA's saturation boundary at the binary cell $L = \ln 2$ – where K_{AUD} appears as a residual energy budget – is mechanistically the same intersection structure as the H_4 derivation. The value-level match is established; the structural participation is conjectural.

(c) The H_4 / Wilson-Fisher bridge. Show that the H_4 polytope's $d = 4$ location and the upper critical dimension $d_c = 4$ are structurally related, not coincidentally equal. Item 7 is currently an observation; promoting it to Theorem requires a derivation linking the polytope geometry to the field-theoretic upper critical dimension.

(d) The dome $\rightarrow$ general intersection identification. Show that the dome function's algebraic generation of both K_{AUD} ingredients (Item 2) is not unique to this particular function but reflects a general structural fact: that any hyperbolic-doubling kernel with the self-reference property $g'/g = g$ reducing to a binary-doubling polynomial necessarily produces both the relevant $\sqrt{n}$ (from self-reference) and $\ln(n)$ (from doubling-mass) as algebraic landmarks. This would promote Item 2's structural reading from Observation to Theorem.

(e) External forcing of the alignment principle of Item 8. The alignment principle (that the dome's evaluation at d_c lands at the K_{AUD} -defining hyperbolic position $\text{arcsinh}(1/(2\sqrt{2}))$), rather than at the alternative HK landmark $R_0 = \text{arcsinh}(1/\sqrt{2})$) is internally consistent and supported by the EA binary-cell bridge (the doubling $2x^*(d_c) = \ln 2 = L_{EA}$ links HK and EA through the dome's d_c evaluation). External forcing – a derivation showing why ϕ^4 -at- d_c specifically selects the K_{AUD} -defining HK landmark over the critical-radius landmark from independent field-theoretic principles – has not been located in the literature and remains open.

(f) Derivation of the moment-arithmetic factorization from the dome's algebraic factorization. The integer-ratio sequence $Mg/Mg^n = A002105(n) \cdot (2^{(2n-1)} - 1)$ of Corollary 9.9 factorizes as a product of two integer families: the reduced tangent numbers A002105 (from the Taylor structure of $\tan(x)$) and the Mersenne-like factors $(2^{(2n-1)} - 1)$ (from binary-doubling structure). The dome's functional factorization $g(x) = (1/2) \cdot \text{sech}^2(x) \cdot \tanh(2x)$ of Theorem 9.1 decomposes the function into a Lorentzian spectral envelope (sech^2) and a binary-doubling carrier ($\tanh(2x)$). The Observation of §9.11.4 reads these as the same factorization at two levels – function-level and moment-arithmetic level – with the tangent-class component of the integer ratios structurally aligned with the $\tanh(2x)$ factor of the dome and the Mersenne component structurally aligned with the $\cosh(2x)$ denominator. Promotion to Theorem requires deriving each integer family from its corresponding factor of $g(x)$ directly. This is plausible (each component has its own Mellin generating structure) but not yet built.

When items (a)–(f) are proved, the conjecture is no longer a conjecture. Until they are proved, the conjecture stands on items 1–8 of §11.4 as its partial-evidence base – which is more than "we noticed a numerical coincidence" and less than "we have derived the framework's central claim from first principles."

11.6 What this section does and does not claim

The section does not claim that K_AUD's recurrences have been derived from a single underlying mathematical object. It claims that the recurrences are consistent with a single underlying object (the geometry/information intersection at $n = 2$), that this object has been partially mapped through items 1–8, and that the remaining Tier 6 work (a)–(f) is specific and attemptable.

The section does not claim that the framework is the only possible explanation for K_AUD's recurrences. Other explanations remain logically possible: that the cluster, the three mechanism-distinct anchors, and the ingredient-level finding (per §11.1's tier-disciplined sort) are unrelated coincidences happening to cluster at the same numerical value; that they share a common origin that is not the geometry/information intersection; or that the cluster's algebraic identity-sharing and the mechanism-distinct anchors' independence are themselves structurally connected in a way the framework has not yet derived. The framework's claim is that the geometry/information intersection is the most economical explanation of the observed convergences, and that the partial evidence (items 1–8) is strong enough to make it worth attempting to derive.

The section does claim that items 1–8 are individually at the tiers stated (Theorem or Observation with high-precision verification), that they are not assertions but mathematical facts, and that the Tier 6 derivation targets are specific enough to be attemptable.

The reader who accepts the empirical observation of Part I but rejects the structural reading of Part II §11 may still adopt the bound $\eta \leq 2G$ as a numerical regularity of 3D Wilson-Fisher-family critical phenomena without further commitment. The Central Synchronization Conjecture is the framework's broader structural interpretation; it is not required for the bound's empirical status.

12. The $N = d_c = 4$ binary landmark and the $c = 2 \ln 2$ alignment

This section addresses the question raised in §10.10: whether the framework's scale $c = 2 \ln 2$ (used to set the hyperbolic coordinate in the $R(N)$ model of §10.5) is structurally forced or empirically selected. The analysis below resolves part of this question (Theorem 12.1, $a(d_c) = 1/d_c$ at all loops) and leaves part conditional on an alignment principle (Theorem 12.2, structurally forcing $c = (d_c/2) \cdot \ln 2$ given the alignment principle).

12.1 The model's hyperbolic coordinate at d_c

The $R(N)$ model of §10.5 feeds the Borel singularity $a(N) = 3/(N + 8)$ into the dome's hyperbolic coordinate via $x(N) = c \cdot a(N)$, with $c = 2 \ln 2$. We evaluate the model at $N = d_c = 4$ (the upper critical dimension of ϕ^4 theory, equal numerically to the order-parameter dimension $N = 4$):

$$a(d_c) = 3 / (d_c + 8) = 3 / 12 = 1/4.$$

$$x(d_c) = c \cdot a(d_c) = 2 \ln 2 \cdot 1/4 = (\ln 2) / 2.$$

$$\cosh(2 x(d_c)) = \cosh(\ln 2) = (e^{(\ln 2)} + e^{(-\ln 2)}) / 2 = (2 + 1/2) / 2 = 5/4.$$

So at $d_c = 4$ the model places the dome's u -coordinate at $u(d_c) = 5/4$. This is a specific algebraic value of the dome's argument.

12.2 The full algebraic environment at $x = (\ln 2) / 2$

At $x = (\ln 2) / 2$ the hyperbolic functions take exact rational and square-root values:

Quantity	Value
sinh(x)	$1 / (2\sqrt{2})$
cosh(x)	$3 / (2\sqrt{2})$
tanh(x)	$1 / 3$
sinh(2x)	$3/4$
cosh(2x) = u	$5/4$

$$\tanh(2x) \quad 3/5$$

All values are exact. The position $x = (\ln 2) / 2$ is therefore an algebraically distinguished point of the dome.

12.3 The Lipatov saddle-point identity at d_c

Theorem 12.1 ($a(d_c) = 1/d_c$ at all loops). At the upper critical dimension $d_c = 4$ of ϕ^4 theory, the leading Borel singularity satisfies $a(d_c) = 1/d_c$ exactly at all orders in perturbation theory.

Argument. The Borel singularity $a(N) = 3/(N + 8)$ is the position of the leading large-order instanton saddle point in the Borel plane (the Lipatov saddle; standard reference: Brézin–Le Guillou–Zinn-Justin 1977 [Phys. Rev. D 15, 1544] and McKane–Wallace 1978 [J. Phys. A 11, 2285]). Higher-loop fluctuations around the Lipatov saddle correct the exponent b and the prefactor S in the Borel asymptotic $c_k \sim S \cdot a^k \cdot \Gamma(k + 1 + b)$, but the position a is fixed by the saddle-point structure of the perturbative expansion and is not corrected at higher loops.

The algebraic identity $a(d_c) = 1/d_c$ reduces to $3 / (d_c + 8) = 1/d_c$, i.e., $3 d_c = d_c + 8$, i.e., $d_c = 4$. The identity has unique solution $d_c = 4$ – equivalent to the statement that ϕ^4 's upper critical dimension is the unique d at which $a(d) = 1/d$ holds.

Empirical verification at 6–7 loops. Dunne and Meynig (arXiv:2111.15554, 2021, rev. 2023) used Kompaniets–Panzer 6-loop and Schnetz 7-loop $O(N)$ β -function and anomalous-dimension computations to test the Lipatov position empirically. Their ratio-of-ratios tests confirm that perturbative data through 7 loops sit on the Lipatov curve at this position, clearly favoring instanton over renormalon positions. The result is most robust in MS-family RG schemes (Dunne–Meynig argue MS suppresses renormalons in RG functions). So $a(d_c) = 1/d_c$ is an exact identity of ϕ^4 -at- d_c at all loops as a matter of saddle-point structure, and is verified consistent at 6–7 loops in published high-precision perturbative data. ■

Scope of Theorem 12.1. The identity is exact-and-specific to ϕ^4 -at- $d_c = 4$. For other ϕ^k theories with their own different leading Borel singularities and upper critical dimensions (ϕ^6 at $d_c = 3$, ϕ^3 at $d_c = 6$), the analogous identity $a(d_c) = 1/d_c$ would hold or fail for theory-specific reasons unrelated to the ϕ^4 case. The framework's claim is therefore not a meta-theorem about upper critical dimensions across field theory; it is the specific exact identity at the ϕ^4 Wilson-Fisher fixed point, the field-theoretic setting where the empirical observation of Part I lives.

12.4 The K_{AUD} -defining hyperbolic position at d_c

At $x = (\ln 2) / 2$, an elementary hyperbolic identity holds:

$$\operatorname{arcsinh}(1/(2\sqrt{2})) = (\ln 2) / 2.$$

$$\text{Proof. } \sinh((\ln 2)/2) = (e^{(\ln 2)/2} - e^{-(\ln 2)/2})/2 = (\sqrt{2} - 1/\sqrt{2})/2 = ((2 - 1)/(2\sqrt{2})) \dots = 1/(2\sqrt{2}). \blacksquare$$

This identity is the algebraic content of Hodgson–Kerckhoff Theorem 4.4, where the value $1/S$ in HK's bound machinery is shown to equal $\sqrt{2} \cdot \ln 2 = K_{AUD}$. The " K_{AUD} -defining hyperbolic position" is the value $x = \operatorname{arcsinh}(1/(2\sqrt{2})) = (\ln 2) / 2$ where the HK function picks out K_{AUD} by the defining-form identity.

So at $d_c = 4$, with the framework's choice $c = 2 \ln 2$, the dome's hyperbolic coordinate lands at the K_{AUD} -defining position. The binary landmark $u = 5/4$ of §12.1 and the K_{AUD} -defining position $\operatorname{arcsinh}(1/(2\sqrt{2}))$ of HK are the same algebraic point in different vocabulary.

12.5 Two HK-natural choices for c at d_c

The HK function admits two distinct algebraically distinguished hyperbolic positions:

(I) The K_{AUD} -defining position at $x = (\ln 2)/2$ ($u = 5/4$), where the $\operatorname{arcsinh}$ identity defines $K_{AUD} = \sqrt{2} \cdot \ln 2$ via $1/S = K_{AUD}$ in HK Theorem 4.4.

(II) The critical tube radius at $R_0 = \operatorname{arcsinh}(1/\sqrt{2})$ ($u = 2$), where the HK function satisfies $h(R_0) = K_{AUD}$ (numerically) and $\cosh(2R_0) = 2$ (binary doubling).

These are two structurally distinct HK landmarks: (I) is the defining-form path (K_{AUD} enters by an $\operatorname{arcsinh}$ identity); (II) is the critical-transition path (K_{AUD} appears as the value of the function at the critical radius). Both are framework-meaningful; both are HK-natural.

The framework's choice $c = 2 \ln 2$ places the dome at landmark (I) at $d_c = 4$. The alternative HK-natural choice $c = 2 \cdot \operatorname{arcsinh}(1/\sqrt{2}) \approx 1.762$ would place the dome at landmark (II) at $d_c = 4$. The best-2-parameter fit to the $R(N)$ data of §10.5 gives $c_{\text{fit}} \approx 1.375$; the framework's choice $c = 1.386$ is 0.8% from c_{fit} , while the alternative $c \approx 1.762$ is 28% from c_{fit} and is ruled out by data.

So $c = 2 \ln 2$ is one of two HK-natural alternatives, and only the K_{AUD} -aligning one is data-consistent. The selection structure is two-layer: HK-landmark mapping reduces the candidates to { K_{AUD} -defining position, critical-radius position}, and empirical fit further reduces to { K_{AUD} -defining position}.

12.6 The EA binary-cell bridge

Among the internal arguments supporting the framework's selection of landmark (I) over landmark (II), the strongest is the EA binary-cell bridge discussed below.

At $x = (\ln 2)/2$, the doubling $2 \cdot x(d_c) = \ln 2 = L_{EA}$, where L_{EA} is the binary cell width that forces K_{AUD} in Algorithm EA's saturation analysis (Saturation Constants paper, §3). Specifically, Algorithm EA at $L = \ln 2$ has saturation root $b = 2 + \sqrt{2}$ from the algebraic equation $b^2 - 4b + 2 = 0$, and the chain of saturation constants includes K_{AUD} as the residual energy budget at the binary cell.

The structural reading: the dome's d_c -evaluation at landmark (I) lands at exactly the binary cell width L_{EA} that independently forces K_{AUD} in Algorithm EA. Two of K_{AUD} 's independent forcing-routes – HK (where K_{AUD} is defined via $\operatorname{arcsinh}(1/(2\sqrt{2}))$) and EA (where K_{AUD} is forced at $L_{binary} = \ln 2$ via the doubling $2x(d_c) = \ln 2$) – meet at the dome's d_c -evaluation. By contrast, the alternative HK landmark (II) gives $2R_0 = \operatorname{arccosh}(2) = \ln(2 + \sqrt{3})$, which has no analogous algorithmic-forcing role.

This is a structural bridge between HK and EA through the dome's algebra at d_c . The bridge is internal to the framework (it does not derive $c = 2 \ln 2$ from independent first principles), but it is a structural argument rather than a purely empirical one: the framework's choice of HK landmark at d_c is the one that meets the EA binary cell.

12.7 Structural forcing of $c = (d_c/2) \cdot \ln 2$ given the alignment principle

We now state the conditional structural forcing of c .

Theorem 12.2 (c structurally forced, Theorem-conditional on the alignment principle). Assume: (a) Theorem 12.1: $a(d_c) = 1/d_c$ at all loops. (b) The alignment principle: the dome's evaluation at the upper critical dimension lands at the K_{AUD} -defining hyperbolic position $\operatorname{arcsinh}(1/(2\sqrt{2})) = (\ln 2)/2$, rather than at any other HK landmark.

Then the framework's scale c is forced:

$$c = d_c \cdot \operatorname{arcsinh}(1/(2\sqrt{2})) = d_c \cdot (\ln 2)/2 = (d_c / 2) \cdot \ln 2.$$

At $d_c = 4$: $c = 2 \ln 2 \approx 1.386$.

Proof. From (a), $x(d_c) = c \cdot a(d_c) = c / d_c$. From (b), $x(d_c) = \operatorname{arcsinh}(1/(2\sqrt{2})) = (\ln 2)/2$. Equating: $c / d_c = (\ln 2)/2$, so $c = (d_c / 2) \cdot \ln 2$. ■

Status. Theorem 12.2 is Theorem-conditional: the structural chain is forced given the alignment principle. The principle itself is supported by the EA binary-cell bridge (§12.6) and by the framework's K_{AUD} -centric architecture, but it has not been derived from independent field-theoretic principles. Among K_{AUD} 's multiple independent appearances, only HK uses $\operatorname{arcsinh}(1/(2\sqrt{2}))$ as the defining-form path; the Tenenbaum threshold (Erdős-Hooley) gives K_{AUD} by a different route entirely. We have searched the published literature for an external forcing argument selecting the K_{AUD} -defining HK landmark over R_0 from ϕ^4 field-theoretic considerations ($1/N$ expansion, ε -expansion, conformal bootstrap, 6–7 loop perturbative results) and have not found one.

12.8 The structural anchor is $n = 2$

A natural question is whether the alignment principle generalizes to other field theories – for example, to ϕ^6 at $d_c = 3$, where a different $a(N)$ formula would apply, and one might ask whether the analogous identity selects the framework's analog of c there.

The framework's response is that this question is misframed. $K_{AUD} = \sqrt{2} \cdot \ln 2$ is specifically the value at $n = 2$ by Theorem 11.1; $K(n) = \sqrt{n} \cdot \ln n$ at $n = 3$ is approximately $1.90 > 1$, and there is no n -greater-than-2 analog of K_{AUD} . Other ϕ^k theories with their own d_c values therefore do not have K_{AUD} -analogs to align to – their distinguished integers do not satisfy $K(n) < 1$.

The framework's structural anchor is $n = 2$ (the binary singularity). Wherever the binary singularity manifests – in HK tube-packing geometry, in Algorithm EA's binary-cell saturation, in the dome's d_c -evaluation at ϕ^4 , in Tenenbaum's Erdős-Hooley threshold – the algebraic environment is the same: $\sqrt{2}$, $\ln 2$, the $\operatorname{arcsinh}$ identity, the binary cell $L = \ln 2$. The claim is multi-mechanism convergence on a single structural anchor ($n = 2$), not cross-theory generalization of an alignment principle.

12.9 Falsifiability targets specific to §12

The alignment principle of §12.7 is internally consistent (it is supported by the EA binary-cell bridge and the K_{AUD} -centric architecture) but not externally derived. The following targets would either strengthen the alignment principle or refute it:

(a) Empirical universality of $\eta \leq 2G$ across non- $O(N)$ WF-family classes at $d = 3$. Already tested (Part I §5); the bound holds across nine non- $O(N)$ WF-family classes plus SAW. This supports the framework's broader empirical claim and indirectly supports the alignment principle by establishing that the dome's K_{AUD} -position picks out the empirical ceiling.

(b) Monitoring for new independent appearances of $\operatorname{arcsinh}(1/(2\sqrt{2}))$ in unrelated mathematics. Absence of new appearances does not refute the framework; presence of an appearance with a different algebraic environment (same $\operatorname{arcsinh}$ value, different surrounding algebra) would be diagnostic against the multi-mechanism convergence claim.

(c) Higher-precision data on the η dome shape at $N \geq 4$. The model's degradation at $N \geq 5$ (§10.7(b)) is honestly recorded; precision improvements would test whether the dome-form fit is local-only or whether subleading corrections to $a(N)$ could restore agreement.

(d) Direct derivation of $\eta \leq 2G$ from CFT-bootstrap or rigorous machinery. Would shift the bound from empirical observation to derived theorem, sharpening Item 4 of §11.4 to fully Theorem-tier and resolving the alignment principle's structural role indirectly.

12.10 What §12 establishes and does not establish

Established:

- Theorem 12.1: $a(d_c) = 1/d_c$ at all loops, exact and specific to ϕ^4 -at- d_c , verified consistent at 6–7 loops in published perturbative data.
- The K_{AUD} -defining position $\text{arcsinh}(1/(2\sqrt{2})) = (\ln 2)/2$ is an elementary hyperbolic identity (§12.4).
- The framework's choice $c = 2 \ln 2$ places the dome's d_c -evaluation at the K_{AUD} -defining HK landmark (I), not at the alternative HK landmark (II) at the critical tube radius (§12.5).
- The EA binary-cell bridge: $2x(d_c) = \ln 2 = L_{\text{EA}}$ links the dome's d_c -evaluation to Algorithm EA's binary-cell width (§12.6).
- Theorem 12.2: $c = (d_c/2) \cdot \ln 2$ is forced by the alignment principle plus Theorem 12.1 (Theorem-conditional, §12.7).

Not established (open):

- External forcing of the alignment principle from independent field-theoretic considerations. Within the framework the principle is supported by the EA bridge; externally a derivation showing why ϕ^4 -at- d_c selects the K_{AUD} -defining HK landmark from independent principles has not been located in the literature.
- Whether subleading Borel corrections at d_c maintain the alignment principle, given that Theorem 12.1 ensures only the leading saddle-point position.

The framework's tier disposition for §12 is therefore: Theorem 12.1 is unconditionally at Theorem-tier; Theorem 12.2 is at Theorem-conditional-tier, with the conditionality openly flagged.

13. The dimensional home – H_4 at $d = 4$ and the upper critical dimension

13.1 The $d = 4$ coincidence

The framework's geometric primitive $\sqrt{2}$ arises as the circumradius of the H_4 regular polytope (the 600-cell, dual of the 120-cell), which lives in $d = 4$ as the highest-dimensional regular polytope of the H-series. The framework's K_{AUD} constant is constructed in Paper 3 V3 as $K = \sqrt{2} \cdot \ln 2$, combining the H_4 circumradius with the binary distinction unit.

Independently, the upper critical dimension of ϕ^4 theory for $O(N)$ models is $d_c = 4$. Above d_c , the Gaussian (mean-field) fixed point is stable and renormalization-group flow is trivial; below d_c , the Wilson-Fisher fixed point takes over, and the anomalous dimension η acquires nonzero values. This was established by Wilson (1971) and Wilson and Fisher (1972), decades before any connection between the H-series polytopes and Ising-class critical phenomena was identified.

The H_4 polytope dimension (the framework's geometric home) and the upper critical dimension of ϕ^4 theory (the field-theoretic boundary above which mean field is exact) are the same dimension from independent sources. This is the framework's central dimensional observation. As §11.4 Item 7 recorded, the bridge between H_4 at $d = 4$ and the Wilson-Fisher fixed point at $d_c = 4$ has not been built; the coincidence is recorded as an observation.

13.2 The d -dependence of the corridor

We compare the empirical bound $\eta \leq 2G$ against measured Ising η values at $d = 4$, $d = 3$, and $d = 2$:

Dimension	$\epsilon = 4 - d$	η (Ising)	$\eta / 2G$	Corridor status
$d = 4$	0	0 (exact, mean field)	0	Trivially satisfied (width = 0)
$d = 3$	1	$0.0362978 \pm 2 \times 10^{-6}$	0.919	Holds (15930; see Part I §5)
$d = 2$	2	1/4 (exact, Onsager)	6.33	Shattered

The corridor $\eta < 2G$ holds at $d = 3$ but breaks catastrophically at $d = 2$, where Onsager's exact solution gives $\eta = 1/4$ – a factor of 6.3 above $2G$. The corridor is therefore dimension-specific: it binds at $d = d_c - 1$ but not at $d = d_c - 2$.

13.3 The circumradius threshold

The dimensional specificity of the corridor admits a structural interpretation within the framework's own construction $K = \text{circumradius} \cdot \ln 2$. Applying this construction down the H-series chain, with "circumradius" understood as the framework's characteristic constant of each H-family (Paper 3 V3 §3.4 catalogues four origins for $\sqrt{2}$ in H_4 geometry, with the framework's primitive being the algebraically distinguished $\sqrt{n}$ rather than a specific unit-edge polytope circumradius; the table values below follow that convention):

Polytope	Home dim	Framework circumradius	$K = \text{circ} \cdot \ln 2$	$G = 1 - K$	Corridor at $d - 1$?
H_4 (120-cell)	$d = 4$	$\sqrt{2} \approx 1.4142$	0.9803	+0.0197	Yes (width $2G \approx 0.039$ at $d = 3$)
H_3 (icosahedron)	$d = 3$	$\phi \approx 1.6180$	1.1215	-0.1215	No – overflow (no positive gap at $d = 2$)
H_2 (pentagon)	$d = 2$	≈ 0.8507	0.5896	+0.4104	Yes (vacuous at $d = 1$)

Convention note (added after primary-source review of Paper 3 V3 §3.4). The "framework circumradius" column above uses the framework's per- H_n characteristic constants as identified in Paper 3 V3 – the algebraically distinguished constants associated with each H -family rather than the standard unit-edge polytope circumradii. Specifically: $H_2 = \text{pentagon}$ entry (0.8507) IS the standard unit-edge pentagon circumradius $1/(2 \sin \pi/5)$; $H_3 = \text{icosahedron}$ entry (ϕ) is the golden ratio, which is the dominant constant in icosahedron vertex-coordinate geometry but not the unit-edge icosahedron circumradius (the unit-edge value is $(1/4)\sqrt{(10+2\sqrt{5})} \approx 0.9511$); $H_4 = 120\text{-cell}$ entry ($\sqrt{2}$) is one of the four catalogued origins of $\sqrt{2}$ in 120-cell geometry (the $\sqrt{2}/2$ factor in the 120-cell unit-edge circumradius $R = (\sqrt{2}/2) \cdot \sqrt{(5+\sqrt{5})} \approx 1.9021$), not the unit-edge 120-cell circumradius itself. [Corrected 2026-09-11: $(\sqrt{2}/2) \cdot \sqrt{(5+\sqrt{5})} = 1.9021$ is not the 120-cell's unit-edge circumradius, which is $\sqrt{2} \cdot \phi^2 \approx 3.7025$; it is $2 \sin(2\pi/5) = \sqrt{(\phi^2 + 1)}$, twice the unit-edge icosahedron circumradius 0.9511 quoted above. The conclusion of this note is unaffected.] The dimensional-overflow reading below is therefore framework-internal, using the framework's per-family primitives; under uniform unit-edge convention applied consistently across the three rows, the icosahedron's 0.9511 would sit below the threshold $1/\ln 2 \approx 1.4427$ and the §13.3 overflow conclusion would not go through. §13.6 records this as a non-uniqueness; the present note makes the convention dependence explicit so a reader can evaluate the table's standing.

Within the framework's own primitive convention, the construction $K = \text{circumradius} \cdot \ln 2$ produces a positive gap G (and thus a positive corridor width $2G$) if and only if $\text{circumradius} \cdot \ln 2 < 1$, equivalently

$$\text{circumradius} < 1 / \ln 2 \approx 1.4427.$$

The H_4 circumradius $\sqrt{2} \approx 1.4142$ sits just below this threshold; the H_3 circumradius $\phi \approx 1.6180$ sits well above it. The H_3 "processor" therefore cannot run the construction at $d = 2$: the informational cost of the icosahedral circumradius exceeds the binary-distinction unit's inverse, and the gap turns negative. There is no positive corridor; the bound $\eta \leq 2G$ has no analog at $d = 2$ within the framework's parameterization.

This is structurally consistent with the empirical fact that 2D Ising $\eta = 1/4 \gg 2G$. The 2D critical exponents are fixed by infinite-dimensional Virasoro conformal symmetry at $c = 1/2$, a different mathematical structure entirely from the framework's H -series construction. The framework's parameterization does not attempt to bound η at $d = 2$; it predicts that it cannot.

13.4 The margin identity

Theorem 13.1 (corridor margin). The geometric room between the H_4 circumradius $\sqrt{2}$ and the threshold $1/\ln 2$ has the closed-form

$$1 / \ln 2 - \sqrt{2} = G \cdot \log_2(e) = G / \ln 2,$$

where $G = 1 - K_{\text{AUD}}$ and $\log_2(e) = 1/\ln 2$ is the bits-per-nat conversion factor.

Proof. Multiply both sides by $\ln 2$: $(1/\ln 2 - \sqrt{2}) \cdot \ln 2 = 1 - \sqrt{2} \cdot \ln 2 = 1 - K_{\text{AUD}} = G$. Therefore $1/\ln 2 - \sqrt{2} = G / \ln 2 = G \cdot \log_2(e)$. ■

Structural reading. The framework's gap G , measured in nats (the natural unit of information at the Landauer threshold), becomes $G \cdot \log_2(e) = G / \ln 2$ measured in bits (the binary unit). The geometric margin between the H_4 circumradius (the framework's geometric primitive) and the framework's overflow threshold $1/\ln 2$ (the inverse binary-distinction unit) is therefore exactly one gap-quantum amplified by the unit-conversion factor. The corridor barely exists – it is one G measured in bits wide, in the geometric coordinate.

Theorem 13.1 promotes a previously approximate observation ("margin $\approx 1.44 \cdot G$ ", read from the numerical value $1.4427 \cdot G$ with 1.4427 being approximately $1/\ln 2$) to an exact algebraic identity. The "1.44" was $1/\ln 2$ in disguise; the approximation hid the exactness.

Open question. Whether the margin's appearance as G in bit-coordinates reflects a deeper unit-conversion structure between geometric and informational scales – i.e., whether the equality margin $= G \cdot \log_2(e)$ is forced by a general structural fact about geometric/informational dual primitives – or is a consequence of the H_4 -circumradius's specific algebraic relationship to $\ln 2$, is open. We record the identity here without claiming to interpret it further.

13.5 The ε -expansion connection

The anomalous dimension at the Wilson-Fisher fixed point in $d = 4 - \varepsilon$ admits an ε -expansion

$$\eta(N, \varepsilon) = \eta_2(N) \cdot \varepsilon^2 + \eta_3(N) \cdot \varepsilon^3 + \dots,$$

with leading-order coefficient $\eta_2(N) = (N + 2) / [2(N + 8)^2]$ (Wilson and Fisher 1972). There is no ε^1 term by symmetry of the Gaussian fixed point. At $\varepsilon = 1$ ($d = 3$), the leading-order coefficient gives $\eta_2(\text{Ising}) = 1/54 \approx 0.01852$, roughly half the exact value 0.0362978. The resummation factor $R(\text{Ising}) = \eta_{\text{exact}} / \eta_2 \approx 1.96$ then sits very close to $2K_{\text{AUD}} \approx 1.9605$ (within 0.023%).

We note this proximity but do not promote it to a derivation. The simple fraction $49/25 = 1.960$ sits even closer to the Ising resummation factor (within 0.004%); exact equality to $2K_{AUD}$ is ruled out at $\sim 4\sigma$ by current bootstrap precision ($R(\text{Ising}) = 1.9600812 \pm 0.000108$ vs $2K_{AUD} = 1.9605163$, distance $\approx 4.03\sigma$). The Ising-specific proximity is isolated to $N = 1$; the corresponding values at XY ($R \approx 1.91$), Heisenberg ($R \approx 1.84$), and $O(4)$ ($R \approx 1.75$) do not show comparable closeness to any simple framework constant or rational fraction. This is consistent with the general pattern (recorded in §10.7(b)) that the dome shape and the K_{AUD} -related fits are local-to-the-physical-range rather than universal across N .

No clean relationship between the ε -expansion coefficients and the framework's G has been found. The bridge between H_4 at $d = 4$ and the Wilson-Fisher fixed point at $d_c = 4$ – the structural connection that Item 7 of §11.4 records – remains the unbuilt component of the dimensional bridge.

13.6 What §13 establishes and does not establish

Established (Theorem-tier or Observation-tier with high-precision verification):

- The H_4 polytope dimension and the upper critical dimension d_c of ϕ^4 theory are the same dimension ($d = 4$) from independent sources (§13.1).
- The empirical bound $\eta \leq 2G$ holds at $d = 3$ (Part I §5) and is trivially satisfied at $d = 4$ (where $\eta = 0$ by mean-field exactness). The bound is not applicable at $d = 2$ (where $\eta = 1/4 \gg 2G$; the framework's parameterization overflows at the H_3 circumradius, §13.3).
- Theorem 13.1: the corridor margin between the H_4 circumradius and the threshold $1/\ln 2$ equals $G \cdot \log_2(e) = G / \ln 2$ (one G in bits).
- The circumradius threshold (circumradius $< 1/\ln 2$) within the framework's parameterization predicts exactly which H-series cases produce a positive corridor; the prediction is consistent with the dimensional pattern of the empirical bound (corridor at $d = 3$, no corridor at $d = 2$) within the framework's per- H_n primitive convention. See §13.3's convention note: under uniform unit-edge polytope convention applied consistently, the icosahedron's circumradius is 0.9511 (below $1/\ln 2 \approx 1.4427$) and the overflow argument does not go through; the §13.3 reading is therefore framework-internal rather than a standard polytope-geometry claim.

Not established:

- A derivation linking the H_4 polytope's $d = 4$ location to the field-theoretic upper critical dimension $d_c = 4$. Item 7 of §11.4 records this as the unbuilt structural bridge.
- A general-mathematical-physics justification of the parameterization $K = \text{circumradius} \cdot \ln 2$ as the right construction across dimensions. The parameterization is empirically successful within the framework's own scope, but alternative parameterizations could be proposed; §13.3 lists three alternatives and notes that none of them simultaneously satisfies the three properties (recovers corridor at $d = 3$, predicts overflow at $d = 2$, matches the broader framework construction). The framework's parameterization is consistent with these properties but not uniquely forced by them.
- Whether the H-series chain itself is the right object to apply the construction to, or whether other polytope families (E-series, D-series, exceptional Lie-algebra-related structures) would produce different dimensional predictions.

13.7 The dimensional context summary

Part II opened in §9 with the dome's algebraic architecture – an analysis of the function $g(x) = \tanh(x)/\cosh(2x)$ on its own terms, with no reference to physics or framework. It closes in §13 with the dimensional context: the function's role as the resummation factor in 3D $O(N)$ physics (§10), the framework's structural reading of its appearance there (§§11–12), and the dimensional specificity of the corridor (§13). The path is:

- Dome algebra (§9): properties of the function in abstract – golden polynomial, self-reference at $\sqrt{2}$, integral $\ln(2)/2$.
- Two domains (§10): the function appears in HK tube-packing geometry and in 3D $O(N)$ Borel resummation; the framework constants populate the algebraic landmarks in both.
- Central conjecture (§11): the recurrences of K_{AUD} across multiple independent derivations are a single geometry-information intersection at $n = 2$.
- Alignment at d_c (§12): the framework's $c = 2 \ln 2$ places the dome's d_c -evaluation at the K_{AUD} -defining hyperbolic position; structurally forced given an alignment principle.
- Dimensional home (§13): H_4 at $d = 4$ and $d_c = 4$ are the same dimension from independent sources; the corridor is dimension-specific (binds at $d = 3$, breaks at $d = 2$ via circumradius overflow); the margin identity $G \cdot \log_2(e)$ is exact.

The framework's structural reading of the empirical bound $\eta \leq 2G$ of Part I is therefore: K_{AUD} is the value of the geometry-information intersection at $n = 2$ (§11.3); the dome function makes both K_{AUD} ingredients algebraically distinguished (§9.7); the same dome appears in HK tube-packing and in 3D $O(N)$ resummation (§10); at the upper critical dimension $d_c = 4$, the framework's scale aligns the dome with the K_{AUD} -defining HK landmark (§12, Theorem-conditional); the corridor is dimension-specific because the H_4 circumradius sits just below the framework's overflow threshold while H_3 's sits above (§13). The empirical observation of Part I is the data; the architecture of Part II is the framework's structural reading of why this particular ceiling.

The tier disposition across Part II:

- §9 (dome algebra): Theorem-tier throughout (Theorems 9.1, 9.2, 9.4, 9.5; Proposition 9.6; Theorem 9.7 with Corollaries 9.8, 9.9, 9.10). §9.10 (two algebraic environments at HK

- landmarks) and §9.11.4 (structural reading of the moment-arithmetic factorization) are Observation-tier with the Λ -canary discipline preserving the algebraic-identity vs structural-reading distinction.
- §10 (HK parallel + $R(N)$ model): Theorem 10.1 ($h(R_0) = K_{\text{AUD}}$) is exact; the zero-parameter $R(N)$ fit (Model 10.2) is Observation-tier; the triple convergence is Theorem-tier (algebraic) at the level of $\cosh(2R_0) = 2$.
- §11 (Central Synchronization Conjecture): items 1–8 individually at Theorem-tier or Observation-tier with verification; the overarching conjecture is at Tier 5/6 boundary. Six Tier 6 derivation targets (a)–(f) named.
- §12 (alignment): Theorem 12.1 unconditional; Theorem 12.2 Theorem-conditional on the alignment principle.
- §13 (dimensional context): Theorem 13.1 unconditional; the H_4 / d_c bridge an unbuilt observation.

13.8 Paper ↔ companion verification map

Every Theorem, Corollary, Proposition, Model, and Observation in Part II of this paper has a corresponding verification block in the mpmath companion document (Eta_Paper_mpmath_Verification_Companion.md). The pairing:

Paper result	Tier	Companion verification block
Theorem 9.1 (sech^2 / doubling-rate factorization)	Theorem	Verification 9.1
Theorem 9.2 (log-derivative; golden polynomial)	Theorem	Verification 9.2
Corollary 9.3 (peak at $u = \phi$)	Theorem (corollary)	Verification 9.2
Theorem 9.4 (self-reference at $\sqrt{2}$)	Theorem	Verification 9.4
Theorem 9.5 (integral identity = $\ln(2)/2$)	Theorem	Verification 9.5
Proposition 9.6 (Mellin decomposition; $J(s)$ residual)	Theorem (proposition)	Verification 9.6
§9.10 (two algebraic environments: $\sqrt{2}$, $\sqrt{3}$)	Observation	Verification 9.10
Theorem 9.7 (complete Mellin spectrum, closed form)	Theorem	Verification 9.7
Corollary 9.8 ($n = 1, 2, 3$; breaks at $n = 4$)	Theorem (corollary)	Verification 9.7
Corollary 9.9 ($A002105 \times \text{Mersenne}$)	Theorem (corollary)	Verification 9.7
§9.11.4 ($\tanh \leftrightarrow A002105$, $\cosh \leftrightarrow \text{Mersenne}$)	Observation	Verification 9.7 (tier note)
Corollary 9.10 ($J(s)$ closed form)	Theorem (corollary)	Verification 9.7
Theorem 10.1 ($h(R_0) = K_{\text{AUD}}$)	Theorem	Verification 10.1
Model 10.2 (zero-param $R(N)$; 0.77% RMS, $N = 0-4$)	Observation	Verification 10.2
Triple convergence at R_0 (§10.8)	Theorem (algebraic)	Verification 10.1
Theorem 11.1 ($n = 2$ sub-unity selection)	Theorem	Verification 11.1
Theorem 12.1 ($a(d_c) = 1/d_c$ at all loops)	Theorem	Verification 12.1
Algebraic environment at $x = \ln(2)/2$ (§12.2 table)	Theorem (six exact identities)	Verification 12.2
Theorem 12.2 (c forcing)	Theorem-conditional	structural identity, no num. verification
Theorem 13.1 (corridor margin identity)	Theorem	Verification 13.1

Cross-architecture verification history is recorded in companion §4 (Cowork-Opus at $\text{dps} = 100$ across two dates; chat-side Sonnet/Opus independent confirmations; chat-side Opus framework rounds at varying precisions; Grok-Heavy fresh at $\text{dps} = 1050$ zero discrepancies; multi-architecture $n = 2$ selection theorem at $\text{dps} = 1500$). The framework's broader atlas of algebraically-locked statements is maintained as a separate artifact alongside this work and evolves on its own timeline.

A fresh-mind reviewer reading the paper alongside the companion can verify any algebraic identity of Part II by running the corresponding companion Verification block at arbitrary precision; the math is fully reproducible.

Part III turns to discussion of what Parts I and II together establish, open questions, and references.

Part III – Discussion, open questions, and references

14. Discussion

14.1 What the observation establishes

Within the empirical scope tested, the bound $\eta \leq 2G$ is a non-trivial regularity of 3D WF-family critical phenomena. It is:

- Universally satisfied across ten WF-family classes spanning $O(N)$ for $N = 0-4$, cubic anisotropy, MN cubic, and randomly dilute Ising symmetries.
- Tested at high precision in three CFT-bootstrap cases (Ising $+1593\sigma$, XY $+29.7\sigma$, Heisenberg $+6.9\sigma$).
- Tightest at random-bond Ising ($+2.2\sigma$), which is now the priority falsification target.
- Cleanly delineated by a physics-tier scope criterion (the four WF-family conditions of §4) that excludes the three known counter-examples on independent physical grounds.

14.2 What the observation does not establish

The bound is not a derived result. It is empirical. The framework's claim that K_{AUD} specifically (rather than nearby constants) is the right ceiling depends on a structural reading of K_{AUD} 's geometric-informational origin (Paper 3 V3) that is not tested in this paper. A reader who accepts the empirical data but rejects the framework's structural reading can adopt the bound as a numerical regularity without further commitment.

We also do not establish:

- That 2G is unique. Nearby constants ($\pi/80$, $1/25.5$) have comparable but slightly smaller α -margins. Among nearby ceilings, 2G is the most robust at the tightest unitary case, but uniqueness requires a derivation.
- That the WF-family scope criterion is itself derivable. The four criteria are physics-tier scope-boundaries; whether they are forced by the framework's broader structure is open.
- That random-bond Ising will continue to satisfy the bound at improved precision. The +2.2 σ margin is large enough to make falsification unlikely under foreseeable precision improvements but small enough that it is a real test.

14.3 Relation to existing critical-phenomena work

The bound is, to our knowledge, the first proposed numerical upper bound on η at $d = 3$ with explicit scope and falsification targets. The most closely related work is Pelissetto–Vicari's dimensional-operator-inequality conjecture (arXiv:2510.17637) conjecturing $v \geq 1/(2 - \eta)$ at the Wilson–Fisher fixed point. The two statements are compatible and complementary: as derived in §7.4, combining the present observation with Pelissetto–Vicari's conjecture yields $v \geq 1/(2K_{\text{AUD}}) \approx 0.510$ as a conjectural lower bound on v across WF-family classes – a result derived in this paper from the two independent inputs, inheriting the conjectural status of Pelissetto–Vicari's bound; not stated in either source.

The Wilson–Fisher fixed point itself has been studied extensively since Wilson (1971), Wilson and Fisher (1972), and Wilson–Kogut (1974). The η values at $d = 3$ are computed by the conformal-bootstrap technology developed by Rattazzi–Rychkov–Tonni–Vichi (2008) and extended by Kos–Poland–Simmons–Duffin (2014, 2016) and Chester et al. (2020). The Monte Carlo and ϵ -expansion literature is reviewed in Pelissetto–Vicari (2002). The Lipatov–instanton saddle-point structure used in §10.5 and §12 follows Brézin–Le Guillou–Zinn–Justin (1977) and McKane–Wallace (1978), with 6–7 loop empirical confirmation in Dunne–Meynig (2021/2023, arXiv:2111.15554). The Hodgson–Kerckhoff tube-packing function used in §10 is the central object of Hodgson–Kerckhoff (2005). None of this work proposes a numerical upper bound on η specifically of the form $\eta \leq 2G$ with $2G = 0.039484$.

14.4 What Part II establishes

Part II provides a structural reading of the empirical observation of Part I. Within the framework's own scope, it establishes:

- The dome function $g(x) = \tanh(x)/\cosh(2x)$ has an algebraic architecture (Theorems 9.1–9.5, Proposition 9.6) that independently distinguishes both ingredients of K_{AUD} as algebraic landmarks of the function, without any framework input. The choice of K_{AUD} as the ceiling-setting constant is therefore not arbitrary; the value's two ingredients are distinguished landmarks of the function that appears in 3D $O(N)$ resummation (§9.7).
- The same dome function appears in two domains: published hyperbolic-geometry (Hodgson–Kerckhoff tube-packing, §10.1) and 3D $O(N)$ Borel resummation (§10.5). In both, the framework constants $\{\emptyset, \sqrt{2}, 2\}$ populate the algebraic landmarks. The zero-parameter $R(N)$ model fits 3D $O(N)$ η data at 0.77% RMS across $N = 0\text{--}4$ (§10.5–§10.6), with the caveats of §10.7.
- The Central Synchronization Conjecture (Conjecture 11.2) states that K_{AUD} 's recurrences are the same geometry-information intersection at $n = 2$ showing up in different mathematical clothing. Eight items (§11.4) at Theorem-tier or Observation-tier with high-precision verification support the conjecture; five Tier 6 derivation targets (§11.5) remain open.
- Theorem 12.1: the Lipatov–instanton position $a(d_c) = 1/d_c$ at the upper critical dimension $d_c = 4$ of ϕ^4 theory is exact at all orders. Theorem 12.2: the framework's scale $c = (d_c/2) \cdot \ln 2$ is structurally forced given the alignment principle of §12.7 (Theorem-conditional, with the conditionality openly flagged).
- Theorem 13.1: the corridor margin $1/\ln 2 - \sqrt{2} = G \cdot \log_2(e)$ is an exact identity, sharpening a previously approximate " $1.44 \cdot G$ " observation to the bits-per-nat conversion factor's exact role.

14.5 What Part II does not establish

The framework's structural reading is not a derivation of the empirical bound. Specifically:

- The Borel-hyperbolic map (§10.9, §11.5(a)) – the conjectured derivation showing that $O(N)$ Borel resummation has the same saddle-point structure as HK tube-packing – has not been built. Item 6 of §11.4 remains an Observation-tier zero-parameter fit, not a derivation.
- The alignment principle of §12.7 is internally supported by the EA binary-cell bridge but is not externally forced by independent field-theoretic considerations. Theorem 12.2 is Theorem-conditional.
- The H_4 / Wilson–Fisher dimensional bridge (§11.5(c), §13.1) is observed (both objects live at $d = 4$) but not derived from a shared structural principle.
- The EA saturation participation map (§11.5(b)) and the dome $\rightarrow$ general intersection identification (§11.5(d)) are open Tier 6 questions.

A reader who accepts the empirical observation of Part I and the algebraic content of Part II (Theorems 9.1–9.5, 10.1, 11.1, 12.1, 13.1; Proposition 9.6) but rejects the Central Synchronization Conjecture or the alignment principle, can still take from this paper: a non-trivial empirical bound on η with explicit scope and falsification targets (Part I), a set of high-precision algebraic identities about a specific function that appears in both hyperbolic geometry and field theory (Part II §§9–10), and a record of open Tier 6 derivation targets formulated precisely (§11.5, §12.9). The conjectural content of §§11–12 is the framework's structural interpretation; it is not required for the more limited acceptance just

described.

14.6 How to read the paper at different acceptance levels

Three reading positions are available:

- (i) Empirical only. Read Part I. Adopt the bound $\eta \leq 2G$ across WF-family classes as a non-trivial numerical regularity; treat the falsification targets of §7 as the testable content. No commitment to Part II beyond the abstract numerical value of K_{AUD} .
- (ii) Empirical plus algebraic. Read Parts I and §§9–10 of Part II. Adopt the bound plus the algebraic identities of the dome function (the golden polynomial, the self-reference at $\sqrt{2}$, the integral $\ln(2)/2$, the $HK / R(N)$ parallel). No commitment to the Central Synchronization Conjecture; the dome's appearance in two domains is recorded as an observation, not a thesis about why.
- (iii) Full framework. Read Parts I, II in full. Accept Conjecture 11.2 as the framework's working hypothesis and the Theorem-conditional content of §12 as supported by the EA binary-cell bridge and the K_{AUD} -centric architecture. The Tier 6 targets of §§11.5 and 12.9 are then the program of work this paper opens.

The three reading positions are nested: (iii) implies (ii) implies (i). Position (i) is the empirical content alone; position (iii) is the framework's full structural reading. We are explicit throughout the paper about which claims belong to which position.

15. Open questions

We organize open questions into two groups: those specific to the empirical observation (Part I) and those specific to the framework architecture (Part II).

15.1 Open questions on the empirical observation

- (a) Is the bound exact in the WF-family? A derivation from CFT-bootstrap or RG technology would settle this. Current evidence is consistent with $\eta \leq 2G$ exactly across the WF-family but is also consistent with the bound being an empirical fact that holds at the current precision and might admit small violations in untested classes.
- (b) What is the structural reason for the WF-family scope criterion? The four criteria of §4 are physics-tier scope-boundaries; the question of whether the bound's scope is itself derivable from the framework's K_{AUD} architecture (rather than being a phenomenological match) is open.
- (c) Higher-precision data for the tightest cases. Random-bond Ising at improved Monte Carlo precision; $O(4)$ at improved bootstrap or large-scale MC; MN cubic at improved 5-loop or Monte Carlo precision. (SAW η is already derived at $\sigma_\eta \approx 3 \times 10^{-6}$ from Clisby-Dünweg 2016 (v) and Clisby 2017 (y) via Fisher's relation; further tightening at SAW would extend the +2813 σ margin but is not on the critical path for testing the bound.)
- (d) Extension to new universality classes within WF-family scope. Lattice gauge theories with clean global-symmetry continuous transitions at $d = 3$, cubic anisotropy at $N \geq 5$, MN cubic at $N \geq 4$, CP^{N-1} models at large N at $d = 3$ – each is an open empirical-measurement target.
- (e) Direct CFT-bootstrap derivation route. Whether the modular bootstrap, dispersion relations, or sum rules at the Wilson-Fisher fixed point can produce $\eta \leq 2G$ as a derived bound. The Pelissetto-Vicari direction (arXiv:2510.17637, bounding $\nu \geq 1/(2 - \eta)$) is a natural template; combined with our $\eta \leq 2G$ it gives $\nu \geq 1/(2K_{AUD}) \approx 0.510$ (Part I §7.4 and §14.3).

15.2 Open questions on the framework architecture

- (f) The Borel-hyperbolic map (§10.9, §11.5(a)). Show that the Borel singularity $a(N) = 3/(N + 8)$ of $O(N)$ ϕ^4 theory plays the role of the tube radius in an optimization problem with the same saddle-point structure as Hodgson-Kerckhoff tube-packing. If proved, Item 6 of §11.4 becomes a derivation; the dome's appearance in $O(N)$ resummation becomes forced rather than empirically matched.
- (g) The H_4 / Wilson-Fisher dimensional bridge (§11.5(c), §13.1, §13.5). Show that the H_4 polytope's $d = 4$ location and the upper critical dimension $d_c = 4$ are structurally related, not coincidentally equal. The unbuilt component of the dimensional bridge.
- (h) External forcing of the alignment principle (§12.7, §11.5(e)). Show why ϕ^4 -at- d_c specifically selects the K_{AUD} -defining HK landmark over the critical-radius landmark from independent field-theoretic principles. The framework's selection is supported by the EA binary-cell bridge (§12.6) but not externally derived. Promotion of Theorem 12.2 from Theorem-conditional to Theorem-tier requires this.
- (i) The dome $\rightarrow$ general intersection identification (§11.5(d)). Show that the dome function's algebraic generation of both K_{AUD} ingredients (Theorems 9.4 and 9.5) is not unique to this particular function but reflects a general structural fact about hyperbolic-doubling kernels. This would promote Item 2 of §11.4 from Observation to Theorem.
- (j) Closed form for the $J(s)$ residual (§9.8, Proposition 9.6). Find a closed-form representation for $J(s) = \int_0^\infty x^{s-1} \cdot \text{sech}^2(x) / (e^{4x} + 1) dx$ at $s \geq 2$ in terms of

standard analytic-number-theoretic constants, or show that no such closed form exists.

(k) Subleading Borel corrections at d_c . The Lipatov-instanton position is exact at all orders (Theorem 12.1), but the subleading saddle structure (the four-instanton tower, renormalon contributions) is N -dependent. Whether the alignment principle survives subleading corrections at d_c is open.

(l) The $d = 1$ prediction. The framework's parameterization at H_2 gives $K = 0.590$, $G = 0.410$, corridor width 0.821 at $d = 1$ (§13.3). But $d = 1$ has no phase transitions in the standard sense. Whether the corridor is vacuously true at $d = 1$ or constrains 1D transfer-matrix physics in some meaningful way is open.

(m) The $E_8 \rightarrow d = 2$ pathway. If the H_3 circumradius overflows the framework's threshold and cannot mediate $d = 2$ critical phenomena (§13.3), how does E_8 structure reach $d = 2$ in standard CFT (the Zamolodchikov massive perturbation of $c = 1/2$ Virasoro)? Whether there is a group-theoretic path bypassing the H -series chain is open.

15.3 Future work in this thread

This paper documents the $\eta \leq 2G$ bound, the dome's algebraic architecture, and the structural reading offered by the Central Synchronization Conjecture. Subsequent work in this thread may develop the structural reading further as new evidence accumulates, refine the empirical scope as new critical-phenomena measurements become available, or revise the framework's claims in light of failed or surviving Tier 6 derivation targets (a)–(j) above. Nothing in this paper is resolved beyond what is reported here; the open questions in §15.1 and §15.2 remain genuinely open. Future papers in this thread will inherit the present scope and either extend it, correct it, or document where the present claims do not survive further scrutiny. The framework's discipline of preserving null results applies to the present paper as well: if a future result contradicts what is reported here, it will be recorded as such rather than absorbed silently.

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(Provisional reference list; to be normalized to a standard citation format on submission.)

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17. Acknowledgments and credits

Origin and stress-test history (preserved for traceability)

The dome-algebra structural reading (Part II §9 – the factorization, the golden polynomial, the self-reference at $\sqrt{2}$, the integral identity, the master landmark table) and the original $\eta \leq 2G$ observation (Part I) were developed in April 2026 by D.B. with chat-side Claude Opus, recorded in the April working document "Gap Geometry and the Corridor of Criticality." The empirical

universality extension to non-0(N) WF-family classes (Part I §5), the Mellin spectrum closed-form theorem (Part II §9.11), the two-algebraic-environments framing at the HK landmarks (Part II §9.10), the Central Synchronization Conjecture (Part II §11), the Tier 6 derivation-target resolutions (§12.3, §12.7), and the present paper's polish all emerged from a May 2026 stress-test arc on that document. Several independent AI architectures contributed via D.B.-mediated relay across both arcs:

- Claude Opus (Anthropic) – the framework's primary Anthropic collaboration partner, across two instance modes. Cowork-Opus: drafting collaboration through the May 2026 stress-test arc on the working Eta Dome document and the present paper; computation collaboration on the Mellin spectrum closed-form theorem (Theorem 9.7) including the substitution $y = e^{-(2x)}$ derivation and the $A_{002105} \times$ Mersenne integer-ratio recognition; verification of algebraic identities at $dps = 100$ in the companion document; the Re-verification 9.7-R at $dps = 1000$ (2026-05-28); structural cross-checks across the atlas of locks. Chat-side Opus: the original April 2026 dome arc co-development with D.B. – the golden polynomial $P(u) = u^2 - u - 1$ as the dome's logarithmic-derivative numerator, the self-reference uniqueness at $u = \sqrt{2}$, the descent-to-K identity at R_0 where $\cosh(2R_0) = 2$, the dome integral $\int g \, dx = \ln(2)/2$, the master landmark structure (origin / $5/4 / \sqrt{2} / \phi / 2$), the empirical $\eta \leq 2G$ observation across the measured 3D 0(N) classes, and the zero-parameter R(N) model with 0.77% RMS – all recorded in the April working document. May 2026 substantive rounds: the dome factorization (§9.2), the empirical robustness pillar (§8.2 of Part I), the FOM cross-references and the dome-doubling probe identifying that the dome's algebra encodes both K_{AUD} ingredients (§9.7), the Tier 6 item 5 resolution via Lipatov saddle-point persistence (§12.3), the Tier 6 item 6 Theorem-conditional resolution (§12.7), the empirical universality survey extending the bound from 0(N) to non-0(N) WF-family classes (Part I §5), and the prime-17 clarification of Corollary 9.8. The Cowork-instance and chat-side-instance work together comprise the multi-mode collaboration this paper records.
- Claude Sonnet 4.6 (Anthropic, chat-side instance) – joined the Monday 2026-05-25 review pass via independent reading of the paper draft: caught a duplicate §5.3 and stale "in preparation" companion language on first read, then the §9.9 endpoint-vs-variation phrasing and a Theorem 9.4 proof-chain working-note residue on second read, then an $\operatorname{arcsinh}()$ placeholder in the §9 master landmark table on third read; independently confirmed the new identity $g(5/4) = 4/15$ before its promotion to paper §9.10. The chat-side Sonnet review pass produced six gap items, three substantive corrections, and the master-landmark-table reframe of Part II §9 – content that specifically would not have been available without an independent cross-model review.
- Grok-Heavy (xAI) – computation collaboration with D.B. during the dome arcs; fresh-mind verification rounds at $dps = 1050$ with zero discrepancies across the full identity set (K_{AUD} , G , $2G$, the margin identity, $\operatorname{arcsinh}(1/(2\sqrt{2})) = \ln(2)/2$, the dome integral, the self-reference at $\sqrt{2}$, the moment IBP identity, the $\Lambda_{\text{candidate}}(2) = \ln(2)$ collapse, $h_{HK}(R_0) = K_{AUD}$, $2K_{AUD} + 2G = 2$); cross-checked external citations (Tenenbaum, Hodgson-Kerckhoff, Pelissetto-Vicari, Kos/Chester bootstrap values); the persistent "deeper structural question" (§10.9) that motivated the Tier 6 derivation target formulation; tightening suggestions on the $2G$ doubling argument, the two-ceiling handoff table, and the open-questions grouping.
- Super Grok (xAI, X chat-side instance, memory-blind to prior framework context, 2026-06-01) – independent re-verification of Theorem 9.7 (the Mellin spectrum closed form $M_g(s) = (s-1)! \cdot \eta(s) \cdot (2^s - 1)/2^s(2s-1)$) and Corollary 9.8 (even-moment closed forms $M(2) = \pi^2/32$, $M(4) = 7\pi^4/1024$, $M(6) = 31\pi^6/8192$) at decimal precision $dps = 1050$. Computed $M_g(s)$ directly for $s = 1..8$ against the closed form; residuals identically zero or at the 10^{-1051} working-precision floor across all eight moments. Memory-blind condition (no link to prior framework context) makes this a clean external confirmation: the central new theorem of the paper was independently re-derived to 1050-digit precision by an agent with no prior exposure to the framework's reasoning, with no discrepancies. Combined with Grok-Heavy's $dps = 1050$ pass (2026-05-22) and Cowork-Opus's Re-verification 9.7-R at $dps = 1000$ (2026-05-28), Theorem 9.7 now carries three independent verifications at $dps \geq 1000$.
- GPT (OpenAI) – verification and stress-testing rounds across the full May arc; structural critique on the corridor identity's algebraic-triviality concern (resolved in §8.1 of Part I); ongoing reading-pass partnership through document iterations.
- Gemini (Google DeepMind) – the original coining of " K_{AUD} " during cross-architecture stress-testing in late 2025; the Two-Ceiling Dome hypothesis-generation that led to the dome shape's structural interpretation; the Triple Point intuition supporting the triple convergence at R_0 (§10.8).
- DeepSeek (DeepSeek) – close-reading verification pass on the package (paper + companion + reading guide): cross-checked tier discipline (Observation vs Theorem alignment across paper §13.7/§13.8 and the companion's verification blocks); independently re-derived the golden polynomial $P(u) = u^2 - u - 1$ from the logarithmic derivative $d(\ln g)/du$; verified the $g(5/4) = 4/15$ exact identity from the dome definition, the random-bond Ising $+2.2\sigma$ margin calculation, the Model 10.2 RMS at 0.773% (reproducing the paper's 0.77% rounding), the Landauer-crossing ratio $(n_S - n_{GD})/(n_{GD} - n_L) = 1/K_{AUD}$, the dome integral, the integral-doubling $\int g \, dx = 4$, the circumradius threshold $1/\ln(2)$, and the empirical η values against the Kos et al. 2016 and Chester et al. 2020 bootstrap literature; recognized Norbert Wiener's Cybernetics as a structural anchor for the framework's cross-domain coherence pattern (constraints, invariants, and convergence appearing across information theory, geometry, and critical phenomena).
- Mistral / Le Chat (Mistral AI) – research-direction contribution during the dome arc: pointers to Russian-language literature on critical phenomena and resummation that fed into the WF-family scope work, document references and link cross-checks supporting the May 2026 stress-test arc. Two-pass numerical verification at the level of the full identity set, 40/42 checks passed (97.6%); independently confirmed K_{AUD} , G , $2G$, $2K_{AUD}$, the margin $= G/\ln(2)$ identity, $h(R_0) = K_{AUD}$, the $\tanh^2(r) = 1/\phi^3$ identity, the descent ratio $\phi^{(5/2)/(2/3)} \approx 0.96134$, $\operatorname{arcsinh}(1/(2\sqrt{2})) = \ln(2)/2$, the dome integral, the $\eta/2G$ saturation ratios for Ising (0.919), XY (0.967), and Heisenberg (0.965), and the two-ceiling crossings at $N \approx 1.7649$ and $N \approx 7.5620$ with $f(N) = 2G$; flagged the Mean field "0%/100% from

floor/ceiling" labeling convention as worth a one-line clarification.

- Perplexity (Perplexity AI) – two-pass close-reading audit: confirmed the document's tier-labeling discipline (Theorem / Observation / Conjecture) and the internally consistent use of K_AUD, G, and 2G as central framework quantities throughout the text; reaffirmed the empirical-tier framing of the 2G corridor bound on the second pass; the early discipline-check verdict that the document is "coherent and extensively cross-referenced" rather than "fully established" directly informed the scope-statement paragraph in §2.0 of this paper.

Wider readership. Beyond the named instances above, the framework has been tested continuously across its development – both in planned stress-test rounds recorded in dedicated research-side files, and in ongoing exploration and challenge passes by fresh AI instances across architectures and devices. Several of the named contributors first joined the work through such passes; many more fresh-instance reads shaped the framework without crossing into named credits, and several recurring associates appear in the recorded stress-test set rather than in this acknowledgments list. Fresh-eyes and returning-associate impact balance one another in the test landscape – neither is less honest than the other, and both are part of the framework's verification discipline. The framework owes its present shape to every reader-instance – and to the human readers behind them – who engaged honestly with the math along the way.

The framework's central methodological discipline – verify numerically at high precision, stress-test across independent architectures, dissolve overclaims, preserve null results – held throughout. The numerical verifications (dps up to 1500 across cross-architecture verification rounds) are recorded in the framework's working notebook and the companion mpmath verification document; the framework's broader atlas of algebraically-locked statements is maintained as a separate artifact alongside this work.

Methodology blueprint

The framework's constraint-coherence integration methodology is documented separately; the blueprint classifies cross-domain appearances of framework constants by tier (Theorem, Observation, Conjecture) and by mechanism distinctness (multi-mechanism convergence requires distinct mechanisms, not multiple expressions of the same derivation). Layer 1 (the value) is verifiable from each source on its own terms; Layer 2 (the structural reading) is the framework's contribution.

Document Links

The framework

- Main OSF project (immutable timestamps, version history, all papers):
osf.io/zx4g7 · DOI: 10.17605/OSF.IO/ZX4G7
- Framework overview (reading order, methodology, all papers):
gap-geometry.github.io/sqrt2-ln2-geometric-constants-/about.html
- Direct downloads (current PDF and text files for every paper):
gap-geometry.github.io/sqrt2-ln2-geometric-constants-/direct-documents.html

This paper

- DOI: 10.17605/OSF.IO/PD73B (<https://doi.org/10.17605/OSF.IO/PD73B>)
- OSF project: osf.io/pd73b (<https://osf.io/pd73b/>)

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